

Micro Motion, Inc

Final Report

Package Drop Tester

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Executive Summary

Team 23, a CU Boulder Mechanical Engineering Senior Design team, was sponsored by Micro Motion, Inc. to design and manufacture a custom onsite package drop tester capable of testing packages up to 90 lbs. Micro Motion currently relies on an off-site testing facility in south Denver that cannot accommodate sustainable packaging alternatives, making in-house testing essential for streamlining their process and exploring eco-friendly options.

The final machine is a free-standing, pneumatically actuated drop tester that supports adjustable drop heights from 12 to 48 inches, and accommodates 10 distinct package orientations (face, corner, and edge) in compliance with ASTM D5276, ISTA 1C/1E, and OSHA safety standards. The system is powered by a single 120V outlet and connects to an external 120 PSI compressed air supply. The machine's core subsystems include:

Drop Mechanism

A pneumatically driven drop leaf assembly with a passive mechanical locking system ensures safe, repeatable package releases. The leaf retracts and resets in under 15 seconds, meeting the reload time requirement for a drop cycle.

Height Adjustment

A chain hoist and linear rail system raises and lowers the carriage with accuracy within the required $\pm 1/0$ inch tolerance, verified across the full 12"–48" range, as well as hold the drop leaf platform within 2 degrees of parallel to the impact surface.

Carriage & Structure

A $\frac{1}{2}$ " bent sheet metal carriage rides along linear rails within a carbon steel column structure, sized to fit within a 3' $\times$ 5' $\times$ 10' footprint.

Electrical System & HMI

An Arduino Nano Every serves as the central controller, managing a state-based software architecture with poka-yoke safeguards including hard and soft E-stops, limit switches, safety relay, and tower status light. A custom handheld controller allows the operator to safely step away from the machine during drops.

All primary engineering requirements were verified and passed, including load capacity, drop height accuracy, package size range, orientation support, reload time, and machine footprint. The total project cost was \$5,039.37 (excluding shipping and labor costs), split between CU Boulder's \$2,000 allotment, with a later extension of \$350, and \$2,689.37 contributed by Micro Motion.

Key lessons learned include the importance of design for manufacturability, tolerance stack-up analysis, and early cost tradeoff evaluation. Key future improvements include pneumatically actuated carriage stabilizers, a digital height readout via distance sensor, and upgraded industrial wire connectors.

The package drop tester has been delivered to Micro Motion's testing facility, providing the company with a safe, repeatable, and fully customizable in-house drop testing solution.

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Summary

Background

Micro Motion, Inc. manufactures and sells Coriolis flow meters, currently being packaged and shipped with toxic and unsustainable packing materials like Instapak Foam. Micro Motion currently tests these packages by drop testing off site at a facility in south Denver but is unable to test the sustainable alternatives due to conflicts of interest. Micro Motion wants to move drop testing on site to streamline their package testing procedures, as well as test more sustainable packaging options. Micro Motion sponsored CU Boulder's Mechanical Engineering Senior Design Team 23 to design and manufacture an onsite package drop tester for packages under 90 lbs. This will allow them to have flexibility with their testing procedures and make continuous improvements to their packaging.

Although there are drop testers already on the market, Micro Motion would like to make multiple modifications and improvements regarding safety and package orientation, to create a more customized testing environment. With this industry and these modifications, standards such as ASTM 5276, ISTA(3A/2A/6/1C), and OSHA safety standards are the most applicable to the project.

Introduction

Team 23 delivered a free-standing functional package drop-testing machine to Micro Motion's testing facility to test current packaging designs and allow for testing up to ASTM standards. There are two primary components to this project: the height adjustment system, allowing for incremental height adjustments to test different packages to their unique ASTM requirements, and the dropping mechanism, allowing for unaffected free fall of packages at 10 specific orientations. Team 23's design incorporates a chain hoist driven height adjustment system for accurate and quick height changes, and a pneumatically actuated drop leaf assembly for repeatable, fast, and accurate package drops. An electronics system with a versatile controller allows for full control over the machine, with poka-yoke mechanisms to prevent operator error and avoid machine damage. Overall, the package drop tester enables efficient and useful package drop testing at Micro Motion Inc.

Project Requirements

Team 23 developed a set of measurable engineering specifications in accordance with the requirements provided by Micro Motion Inc. These specifications were established to ensure that the final design satisfies all functional, safety, and performance criteria outlined by Micro Motion. The client's top priority was user safety, along with compliance with the testing conditions specified in Micro Motion Inc.'s standard drop test protocols. Table 1 below summarizes the key engineering specifications, their corresponding requirement values, the verification methods used to ensure that the design meets Micro Motion Inc.'s established criteria, and the results of verification testing.

Table 1: Project requirements with verification methods (revision 2).

ID	Category	Specification Statement	Status	Industry Standards or Codes (or N/A)	Verification Type	Verification Results
1	Power	The device shall be powered by a single 120V outlet	Firm	N/A	Demonstration	Passed
2	Drop Height	The device shall support adjustable drop heights ranging from 12 inches to 48 inches, in 2-inch increments, with a tolerance of +1 inch	Firm	N/A	Demonstration	Passed
3.1	Load Capacity	The device shall support a maximum weight of 90 lbs	Firm	N/A	Test	Passed
3.2	Package Size	The device shall operate as intended at both the maximum and minimum package sizes	Firm	N/A	Demonstration	Passed
3.2.1	Longest Package Size	The device shall support packages no longer than 40" x 43" x 16".	Firm	N/A	Demonstration	Passed
3.2.2	Smallest Package Size	The device shall support packages no smaller than 12" x 9" x 13"	Firm	N/A	Demonstration	Passed
3.3	Package Orientation	The box holding mechanism shall support the 10 desired orientations (face, corner, edge)	Firm	ER-3100412 Section 3.3	Demonstration	Passed
4.1	Product Size	The entire machine shall not exceed 3' x 5' x 10'	Firm	N/A	Inspection	Passed
5.1	Reload Time	The system shall perform a complete drop and reset cycle in ≤ 15 seconds	Preferred	N/A	Demonstration	Passed

Final Design Overview

Drop Mechanism

Leaf Structure

The platform that supports the package before it is dropped is referred to in industry as the “drop leaf,” and the team has adopted the same convention. The base requirements for this structure are (1) to support the weight of all specified packages and (2) retract downwards and out of the way of the package, so that it is allowed to free fall onto the impact surface.

Several secondary requirements are derived from other aspects of this function. The most significant is the need for the platform to support a package with a largest face of 40” x 43”, requiring at least 20” between the center of mass of the package and the machine, combined with the minimum drop height of 12”. It would be impossible for a platform to simply rotate downward to a vertical position within those constraints, so a more complex approach is required. Team 23’s solution is to add a vertical offset to the pivot point of the assembly, reducing the height of the arc swept by the platform.

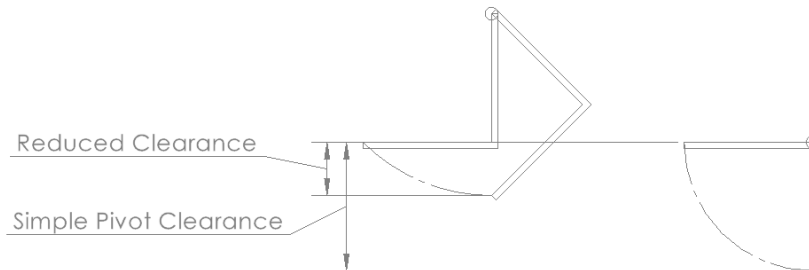


Figure 1: An illustration of the effective height reduction achieved by offsetting the pivot to an axis above the platform.

The drop leaf is primarily composed of four steel ribs, each half an inch thick. They are bolted together in pairs with spacers, and plates bolt to the spacers on top of the leaf to secure the entire assembly together. Two of these plates are 2” x 8” and positioned near the ends of the vertical part of the leaf to increase the rigidity of the assembly. Another plate is 8” x 10” and is placed at the end of the horizontal part of the leaf. This forms a stable platform where packages can be loaded.

The assembly has been designed to accommodate packages of up to 90 lbs and was validated with FEA to a safety factor greater than 6. The primary structural concern in the assembly was the bending loads induced by holding the package far from the machine structure. Weight was also important, as this assembly is required to move at high speed. These constraints drove the use of ribs, as these proved to provide the best bending resistance per unit weight while keeping the structure simple and manufacturable. The horizontal part of the structure also tapers to a much smaller cross-section near the end to save weight.



Figure 2: A render of the drop leaf assembly.

The drop leaf assembly is connected to the main structure of the machine in two places. First, a threaded block connects the shaft at the top of the assembly to a pneumatic piston, and second, two pairs of rollers sit in a slot in the leaf carrier assembly to guide the leaf along the correct path during the drop sequence. The connection block is 1" x 1" x 2" with a threaded hole in the center of its top face. The block has a 5/8" hole that the top drop leaf axle goes through. This connection is vital for the transfer of energy from the pneumatic piston to the drop leaf.

The force required to accelerate the drop leaf faster than gravity and safely clear the package's path was calculated to be 300 lbs. Due to the significant force at the center of the top axle, analysis was done to ensure the top axle could withstand it. With an axle made of 1144 steel, the axle can safely withstand the force of the piston with a significantly large safety factor of 8. The axles are 0.625" in diameter and pass through all four of the drop leaf ribs and a bearing at each end. Further analysis was done on the ends of the axles for the moment the lower bearing impacts the curve in the leaf carrier track. The ends of the axles experience a higher stress concentration; however, they can withstand the stress of impact with a safety factor of 2.

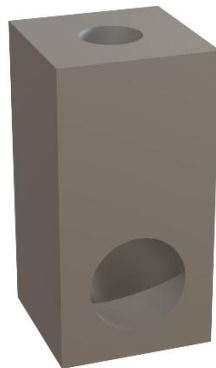


Figure 3: A render of drop leaf connection block.



Figure 4: A render of the axle assembly.

Steel radial bearings are used to allow the axles to move along the track plate path. These are held in place by retaining clips on both sides of each bearing. The main constraining scenarios were the static case of holding a 90 lbs package before a drop and the instant that the bottom roller impacts the curve of the track. In these scenarios, these forces were located primarily on the bottom rollers. The steel bearings have a static load capacity of 747 lbs and a dynamic load capacity of 1,525 lbs, confidently supporting the motion of the drop leaf.

Carrier Assembly

The function of the drop leaf carrier assembly is to provide a track for the leaf to ride in during the drop sequence. The structure consists of two machined steel plates bolted to the bottom of the carriage assembly and secured to each other with aluminum spacers. The slots for the rollers go all the way through the plate, with separate thin plates bolted to both sides of both plates to act as guides and keep the leaf from moving sideways on the track.

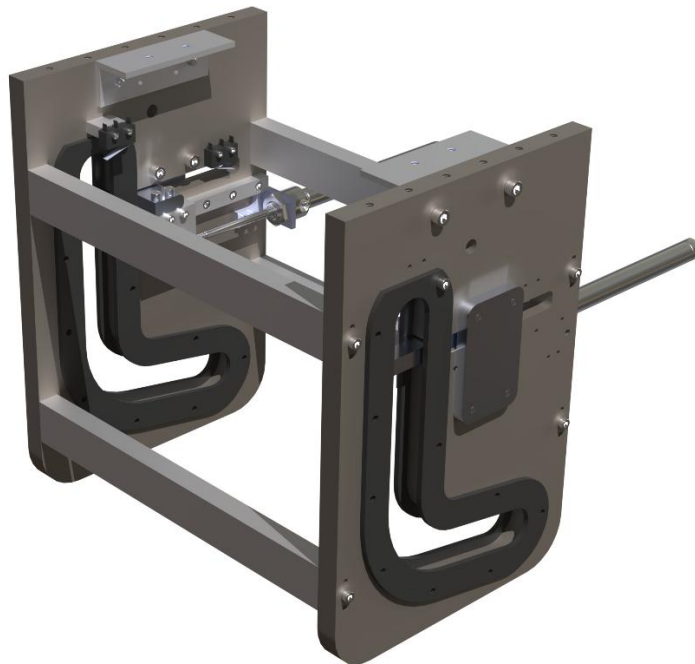


Figure 5: A render of the assembled track plates. In this orientation, the package would sit on the drop leaf to the left.

The strength, rigidity, and precise alignment of this assembly were critical to the function of the drop mechanism. The choice of 0.5" steel plates was driven by the desire to maximize rigidity of the track, minimize wear, and be close to the thickness of the bearings that would ride in the slots. Having a single large component for each half of the assembly also helped keep position tolerances tight. Despite the high stiffness of these plates, the slot for the locking mechanism running through most of its width became an area of concern. The high dynamic loads of the drop sequence created some undesirable deflection modes caused by the stress concentrations around the ends of the slot. To mitigate these concerns, an aluminum brace was placed across the slot near its center to constrain the lower portion of the plate.

Keeping the two track plates aligned to each other was of particular importance, so the four aluminum spacers between them each included holes for a pair of alignment pins at each end. These hardened steel pins are press fit into the track plates and hold them to a position deviation no larger than 0.004".

The guide plates on the outside of each slot were made of Delrin, a plastic material with low sliding friction on its surface. These would experience minimal loading, but keeping friction low was a large consideration to prevent the assembly from binding and to minimize wear. Other mechanical features include angle brackets to help hold the track plates perpendicular to the carriage and parallel to each other, and three limit switches to track the position of the drop leaf and locking system and provide feedback to the control system.

The path of the slot in these plates is also crucial to the drop sequence. The top roller is constrained to move vertically with the pneumatic cylinder that drives it, and the bottom roller controls the rotation of the leaf as it moves downward. The team conducted several stages of dynamic analysis to settle on the current configuration, which is designed to allow the leaf to drop straight down to create separation from the package, then rotate backwards out of its way. The combination of these two motions made analysis a challenge, but the advantage of this system was clear: the initial linear downward motion allowed the leaf to separate from the package most quickly, reducing the force required to achieve a successful drop by about a factor of four compared to a simple fixed pivot point.

The leaf was initially modeled by assuming constant linear and angular accelerations and using kinematic equations to calculate its position and rotation at one-millisecond intervals. This set of equations, after a set of coordinate transformations, could be compared to an object falling by gravity. The results of these calculations can be seen in Figure 6.

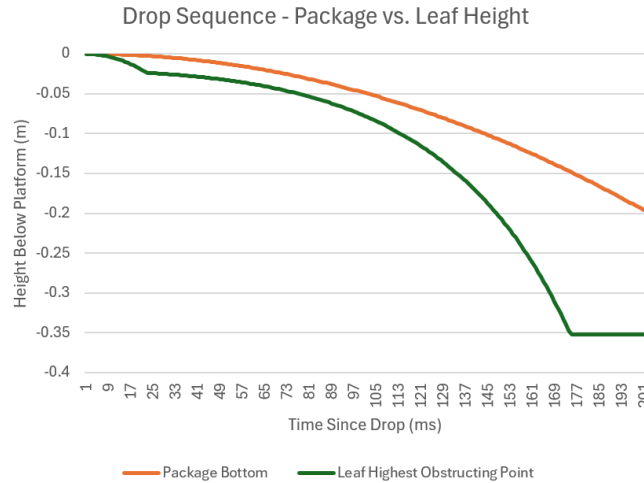


Figure 6: A plot of the highest point on the leaf compared to the predicted height of the package. It demonstrates that the leaf will not collide with the package under these conditions.

Further modeling was performed using physics-based dynamic simulations to capture the full complexity of the drop sequence. These simulations were used to refine the motion path and included additional factors such as rolling friction and the weight of the package. The results allowed the mechanism to be finalized and validated, and also fed into the final specifications for the speed and force the pneumatic system had to be designed for, informing actuator and system design choices.

Locking Mechanism

The default (rest) position of the drop leaf, under a depressurized pneumatic system, is the retracted (down) position. Maintaining the leaf in the upright position using pneumatic pressure alone was deemed undesirable due to both efficiency and safety concerns. Relying on sustained air pressure during operation introduces risk, as a loss of pressure, such as during an emergency stop, would result in unintended motion. During an emergency stop condition, power is removed from the system, causing the control valves to vent and depressurize all actuators.

To mitigate these risks, a passive mechanical locking system was implemented to secure the drop leaf in the extended position. This system consists of a pair of aluminum locking tabs that translate beneath the top rollers, physically preventing downward motion of the leaf. The tabs are actuated by single-acting pneumatic cylinders (7/16" bore, 2" stroke) operating at 100 psi, providing approximately 15 lbf of retraction force. These cylinders are configured as air-retract, spring-extend, such that the default (unpressurized) state corresponds to the tabs being fully extended and engaged.

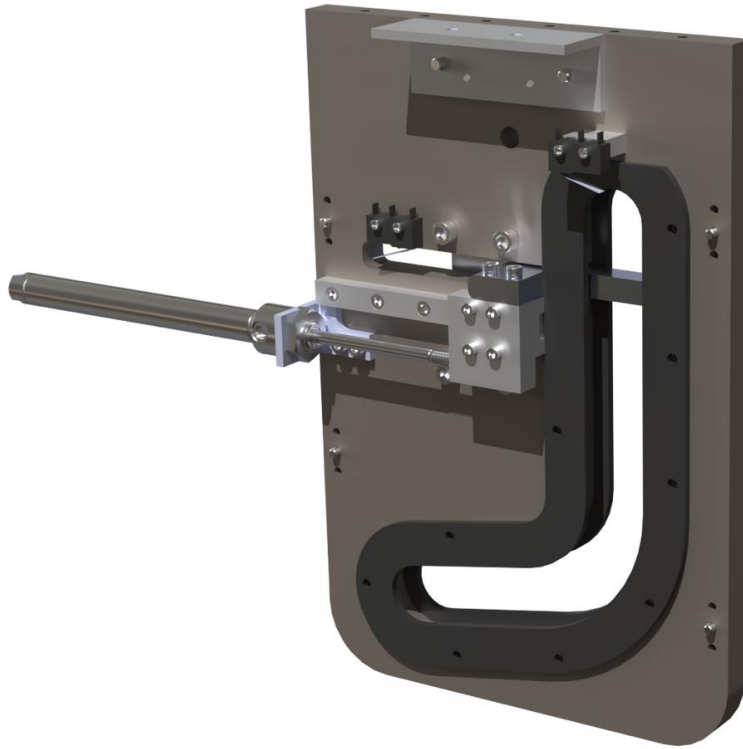


Figure 7: A render of one of the carrier plates with the locking mechanism installed, shown in the locked position.

Each locking tab is manufactured from 6061 aluminum and features a rectangular engagement section measuring 0.325" in height, 0.4" in width, and 3" in length. The tab spans a 0.139" gap beneath the roller and is supported on both ends, forming a simply supported beam under load. This geometry allows the tab to effectively resist both bending and shear forces induced by the supported load.

When engaged, the locking tabs support the combined load of the platform weight (approximately 60 lbf) and the applied force from the main pneumatic cylinder (up to 300 lbf), resulting in a total worst-case loading condition of approximately 360 lbf. Structural analysis of the tabs was performed using FEA, evaluating both bending and shear stresses under this loading scenario. The results indicate a factor of safety slightly greater than 4, confirming adequate strength for the intended application.

An additional safety feature of the design is the inherent binding behavior of the tabs under load. As force is applied downward through the rollers, the tabs experience slight deformation and contact forces that increase friction within the guide interface, preventing retraction while loaded. This ensures that the locking mechanism cannot disengage until the load is first removed. Retraction of the tabs occurs only during the arming sequence, when the main cylinder is pressurized to lift the leaf and unload the locking interface.

The actuation sequence is controlled to ensure proper engagement and disengagement. During locking, the main cylinder first raises the leaf, after which the tabs extend into position.

The load is then transferred onto the tabs as the system depressurizes. The unlocking sequence proceeds in reverse: the main cylinder re-engages to lift the leaf and remove load from the tabs, allowing them to retract.

System state verification is achieved using limit switches. One limit switch confirms that the locking tabs are fully retracted prior to initiating a drop, preventing interference. A second limit switch verifies that the drop leaf has reached the correct position after depressurization. If the locking tabs fail to extend properly, the drop leaf will not remain elevated, and the corresponding limit switch will not trigger, providing indirect confirmation of improper lock engagement.

Material selection for the locking tabs favored 6061 aluminum due to its strength-to-weight ratio, machinability, and sufficient stiffness for the given loading conditions. Although steel would provide higher absolute strength, the analysis indicated comparable bending performance for this application, making aluminum a more efficient choice.

The locking tabs are guided using commercially available linear rails, allowing smooth actuation with minimal friction during normal operation. Notably, the tabs do not contact the rail surfaces under standard conditions; contact occurs only during depressurization, when slight downward displacement introduces load into the system. While long-term wear was not a primary design focus, this limited contact reduces friction and wear during repeated cycling.

Overall, this mechanical locking system provides a robust, fail-safe method of constraining the drop leaf in the extended position, eliminating reliance on pneumatic pressure for load support and significantly improving system safety during both normal operation and emergency conditions.

Damping

The drop leaf needed to stop at a 45-degree angle to avoid package interference. To ensure safe machine operation and prevent machine failure, Team 23 created a dampening system to soften the drop leaf stop. Using a dynamic model of the drop leaf's motion, the kinetic energy of the drop leaf at its final position was estimated to be 2,600 to 3,500 in-lbs. In addition to damping this energy, the damping system could not affect the drop leaf's motion until after the package had cleared, which limited the applications of any attached shock absorber. The final speed calculated at the end of the drop leaf's path was estimated to be 5 m/s. This impact velocity further restricted applicable shock absorbers which had a maximum impact speed of 2 m/s. To fit the requirements of the damping system, 6 polyester elastomer dampers were used. After testing various dampers with impacts, profile dampers with an energy absorption of 50 in-lbs and a stroke length of 1" were selected. This allowed the dampers to be fully compressed with less kickback of the drop leaf.



Figure 8: A render of the damping mechanism assembly.

These 6 dampers are mounted using M6 bolts, 4 of which are mounted to the front steel plate, the other 2 are mounted to the rear steel plate. The front plate is orientated at a 45-degree angle to ensure the 4 dampers impact the drop leaf forks. The front and rear steel plates are connected via two steel connectors and two 10-32 bolts at both ends of each connection. The damping mechanism is attached to the carriage via the two steel hinges on the top, which allows the mechanism to pivot and impact the structure. The structure is impacted by the two dampers mounted to the rear plate. ½” thick steel plates were chosen as they are subject to repeated impacts from the drop leaf. The long steel plates that connect the rear damping plate and the steel hinges keep the damping mechanism fixed and oriented correctly for impact with the drop leaf.

Pneumatic System

The choice to use a pneumatic system on the machine was largely driven by the high power requirements of electrical actuation systems. Given the requirement for a very brief application of large force for each drop, using a pneumatic system to store energy in the form of compressed air decreased the power requirement dramatically. It was also desirable to use pneumatics because air cylinders can provide very powerful actuation in a relatively small package. The production facility where the machine is to be used has a 120 psi compressed air supply available from an industrial compressor and air preparation system, which made the design of the compressed air system on the machine relatively simple.

The pneumatic system has a small manifold on the electrical box which receives air from the external supply. Two hand valves allow the supply to be shut off and vent the machine’s internal system to atmosphere, allowing the machine to be quickly made safe for shutdown or maintenance.

From the manifold, the air runs through a small tube around the side of the frame and up to the carriage through the same cable chain that carries the electrical wiring. This tube was a critical flow restriction to design around, as it is about eight feet long. The bend radius inside the

cable chain restricted the line to 0.25" OD tubing. This created a design conflict, as the drop cylinder requires a very high flow rate of air with minimal pressure drop, and this long tube created an unacceptable pressure drop. Under the conditions of a drop, the pressure at the end of this line dropped as low as 50 psi. The solution to this issue was to include a 0.5 gallon air tank on the carriage, allowing the drop cylinder to pull high-pressure air across a relatively short run of tube. The system is set up to create the absolute minimum restriction between the tank and the down side of the drop cylinder by using larger diameter 0.375" OD tube. The tank is then refilled by an adjustable regulator to 100psi.

The other important high-speed air flow is from the up side of the drop cylinder through its valve to atmosphere, so this connection is also made from the larger tube size. This needs to empty quickly because this side of the cylinder holds the drop leaf up just before the drop sequence after the locking mechanism disengages. If it emptied slowly, the pressure left in it would oppose the drop and reduce its speed.

The remaining parts of the system are for the up supply to the drop cylinder and the locking cylinders. These do not require high flow rates, so they use the smaller 0.25" tubes and all split from one connection. The up input for the drop cylinder controls the reset sequence, which was specified to be slow relative to the drop, so an additional flow restriction valve was included to cause the cylinder to retract slowly—at about a fifth of the speed of the drop. The locking cylinders have a similar flow restriction valve to control the speed of their motion and reduce wear on their components.

There are three electronically actuated valves in the system. One is for the locking cylinders, and the other two are for the up and down sides of the drop cylinder. All three are identical internal-pilot solenoid valves selected for their low cost and fast actuation. They are driven by a 12V signal from the electrical system and were tested to fully change position in less than 50ms. This fast response speed was important for correctly timing the drop sequence, but is also desirable for safety. An emergency stop input will cut power to all three valves, and it's critical that this immediately depressurize all actuators and lock the drop leaf in place.

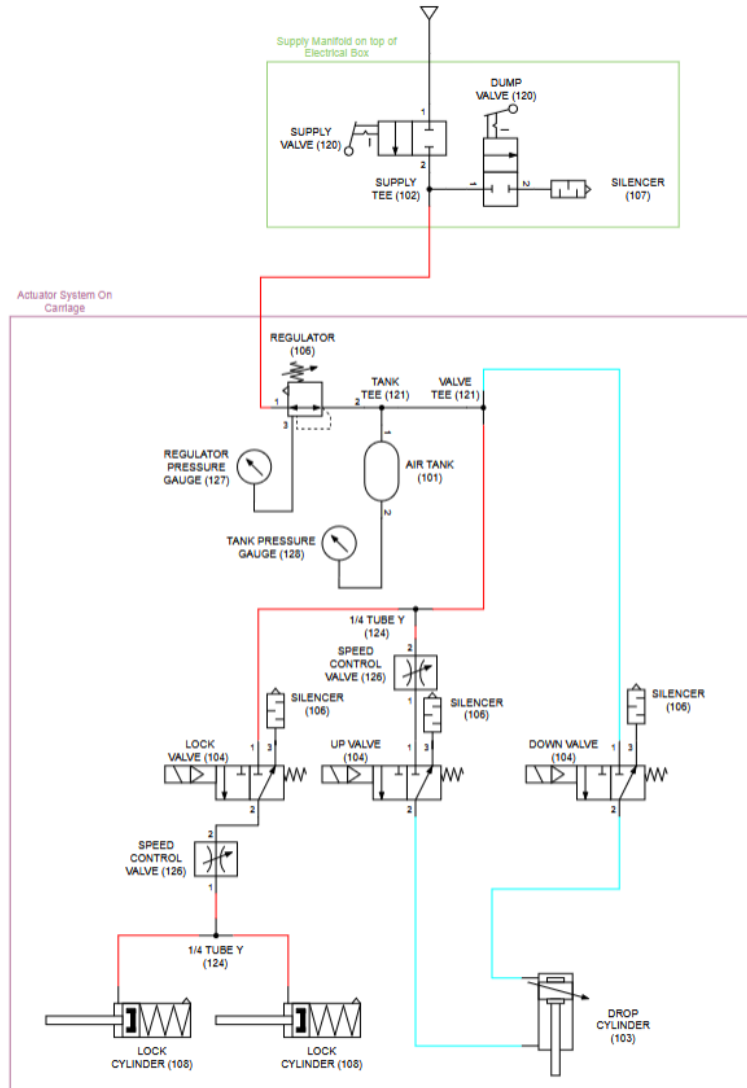


Figure 9: A diagram of the pneumatic system, with general air flow from top to bottom. The locking cylinders are at the bottom left, with the drop cylinder at the bottom right.

Iterations

In a previous version of the drop leaf design, an important constraint was that the leaf had to not interfere with a lead screw which ran up the center of the machine. In this design, the lead screw was directly behind the drop leaf, so the structure needed to be split into two parts and allow the lead screw to pass between them. The influence of this constraint remained and drove the drop leaf to be primarily composed of four steel ribs, each half an inch thick.

Carriage

The carriage is designed to house and support the drop mechanism and associated subsystems. It is fabricated from ½” thick A36 bent sheet steel, with overall dimensions of 18” in width, 21” in length, and 5” in height. This configuration was selected to ensure structural rigidity while maintaining manufacturability. The bending tolerance of $\pm 0.5^\circ$ results in a maximum deviation of approximately 0.0436” over the 5” height, which is well within acceptable limits for maintaining perpendicularity.

At the center of the carriage, a chain hoist hook is mounted and positioned near the system’s center of mass. This placement ensures stable and controlled vertical motion during height adjustments. Since this hook experiences the primary loading during lifting, FEA was conducted to validate the structural integrity of the carriage. The maximum loading condition occurs during upward motion and corresponds to the combined weight of the carriage and its attached components, approximately 133 lbs. Under this load, the carriage exhibits a maximum deflection of 0.0162” and achieves a factor of safety of 4, confirming adequate structural performance.

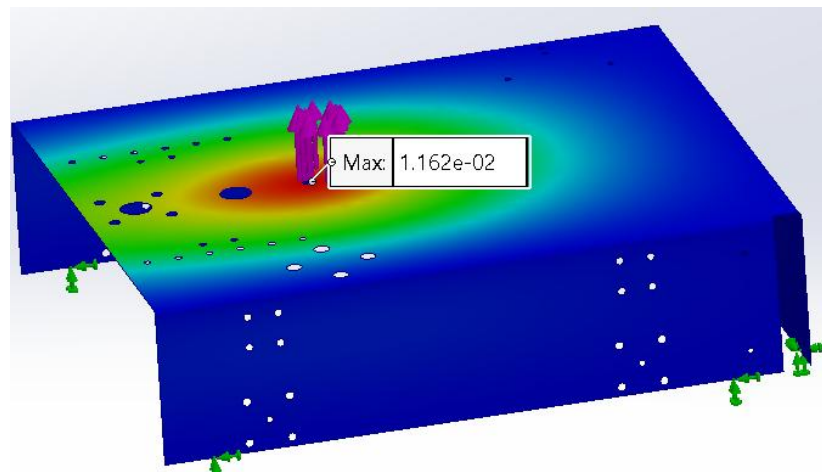


Figure 10: Finite element analysis on carriage.

During manufacturing and assembly, a binding issue was observed that prevented smooth vertical motion of the carriage along the structure. This was traced to slight misalignment caused by structural tolerances. To resolve the issue, slots were introduced in the linear bearing mounting points on the side of the carriage most affected by misalignment (the right side of the machine). These slots, shown in Figure 11, allow for minor positional adjustments during assembly. This modification successfully eliminated binding and enabled the carriage to travel freely across the full required height range of 12” to 48”.



Figure 11: Final design of carriage, with linear bearing slots.

Package Orientation Fixture

To hold packages on an edge or at an angle for specific drop testing cases, the machine required a box angle fixation assembly, shown in Figure 12 below. This assembly consists of three linear steel rods; two are 3' long and the other is 6" long. Two shaft connectors hold the shafts at adjustable angles, with the smaller shaft bolting directly to the box angle fixation claw, shown in Figure 13. The entire angle fixation assembly was designed for ease of use and to meet the package drop orientation requirements.

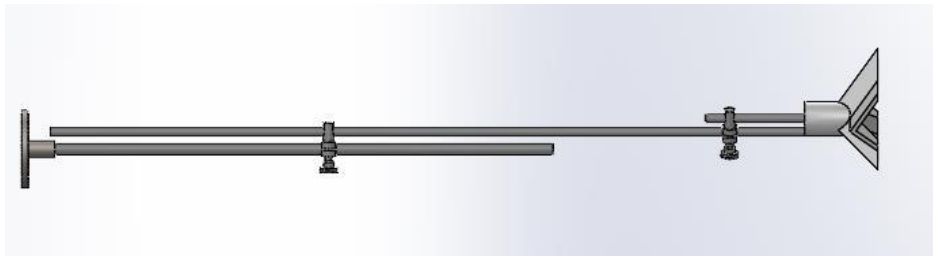


Figure 12: Box angle fixation assembly.

Due to the large range of package sizes and weights, the design originally included two identical angle fixation assemblies. After iterating and testing different designs, it was determined that only one was needed.

The last part of the assembly is the box claw which contacts the package and holds it in the desired orientation. The claw is designed so that it can hold packages by both corners and edges. For ease of manufacturability, the claw was 3D-printed using a SLS printer for strength.

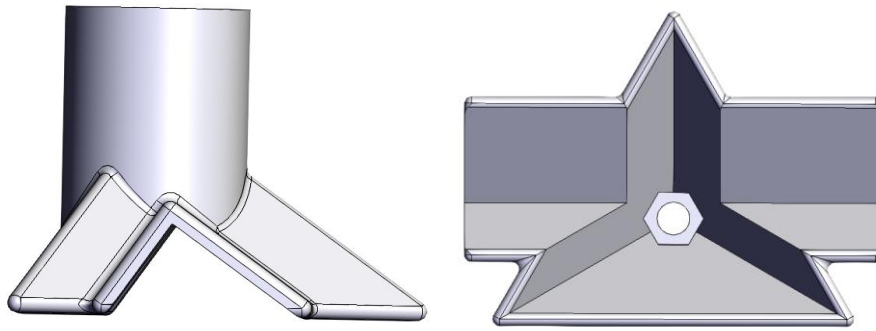


Figure 13: Box angle fixation claw.

Carriage Stabilizers

After analyzing the dynamics of the drop, it was determined that the upward force exerted onto the carriage by the leaf was about twice as much as the combined weight of the carriage and its attached subassemblies. This would have caused the carriage to jump up during the drop sequence and would have impacted the quality of the test. To counteract this effect, carriage stabilizers were added on each side of the carriage. These stabilizers clamp directly onto the structure and ensure the quality of the drop.

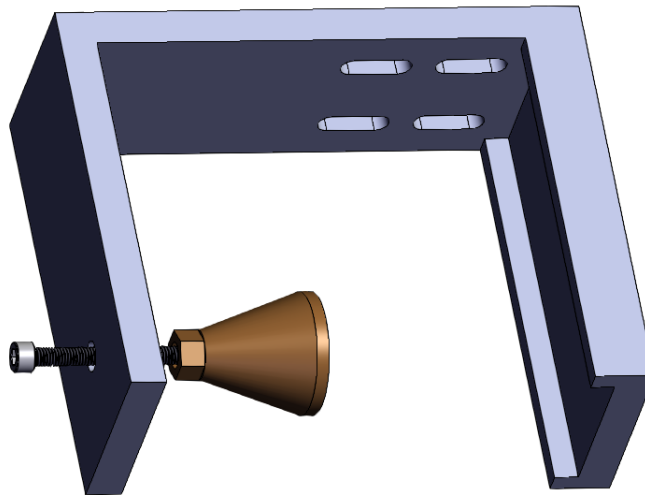


Figure 14. Carriage stabilizer.

Because these stabilizers are purely mechanical, the control system has no way to tell their current state. The user is prompted to tighten and loosen the stabilizers, but they were designed to not restrict the motion of the chain hoist in the case of incorrect operation.

Iterations

There were several key adjustments to the carriage throughout the design process. The first design utilized wheels that clamped onto v-groove tracks on the structure, shown below in Figure 15.

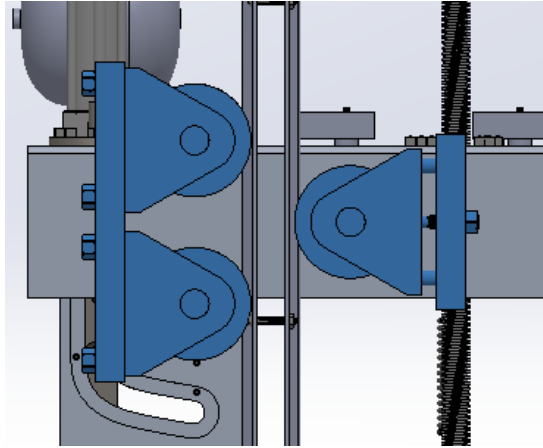


Figure 15: Old carriage design with wheel assembly configuration.

Most of the on-board pneumatic hardware was also removed from the carriage after the team was informed that external compressed air would be available. This change allowed the carriage to be made smaller and lighter. Its dimensions were reduced by 4" in depth and 2" in width, while the height remained unchanged.

With the new 0.5 gallon air tank, a clearance issue unfolded which prevented the system from reaching the 48" drop height. The air tank was relocated to the front of the carriage to solve this clearance constraint. This forced the removal of one of the angle fixation arms, but it was found that one angle fixation arm was enough. After the second redesign phase after Winter break, the height of the structure was increased due to the addition of the chain hoist system, allowing the air tank to move to the back once more, with 1 angle fixation arm deemed sufficient to hold the boxes in position.

Another carriage design that was explored was using steel plates that were bolted together instead of the bent sheet metal. This option was quickly ruled out, as large gussets would need to be added to support the weight of the steel plates. This created clearance issues in the drop mechanism housed by the carriage, as seen in Figure 16. The other issue the steel plates created was that, due to their size, they would need to be outsourced as they were too big for the machines in the machine shop.

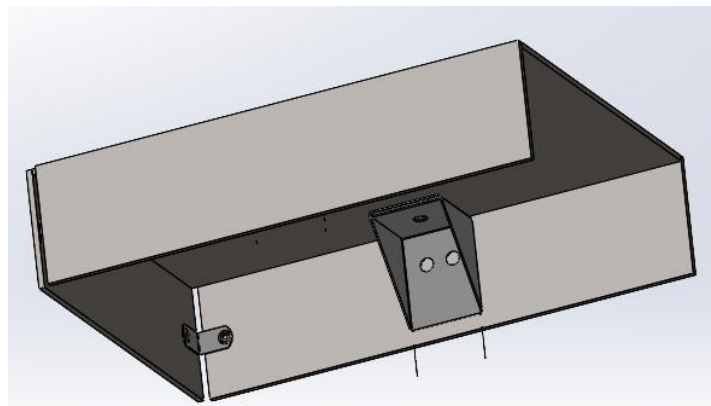


Figure 16: Bolted steel carriage design.

Height Adjustment and Structure

Height Adjustment and Structure Overview

The height adjustment and structure subassembly houses all other components and serves as mounting point for the electronics and other subsystems, while also being structurally sound to take the impact forces because of a drop sequence. The components that make up this subassembly include the main structure, the impact surface, the chain hoist assembly, and the linear rails.

Chain Hoist

The primary height adjustment mechanism is an off-the-shelf chain hoist from ProWinch, rated for 660 lbs. As seen in Figure 17, the chain hoist is connected to the main structure and carriage via hoist hooks. These hooks connect to a hoist ring on the carriage, as well as a chain hoist mount, which is a static version of a purchasable chain hoist trolley mount.



Figure 17: Chain hoist assembly.

The chain hoist mount consists of two 3/8" steel plates connected via a 1" steel clevis pin. On the clevis pin, there is a tab with two machined holes. This tab allows for rotation about the pin to remove the necessity for keeping high tolerances during assembly and welding and allow for slight adjustments along the "x" axis of the machine. Additionally, between the two plates, there are 24 1" washers to allow for incremental adjustments along the "y" axis of the machine. This allows the chain hoist to be placed directly above its connection on the carriage and makes the motion of the chain hoist truly vertical. In Figure 18, the FEA analysis with the loading scenarios present on the machine can be seen, indicating a factor of safety of 3 and 19 for the tab and plates, respectively.

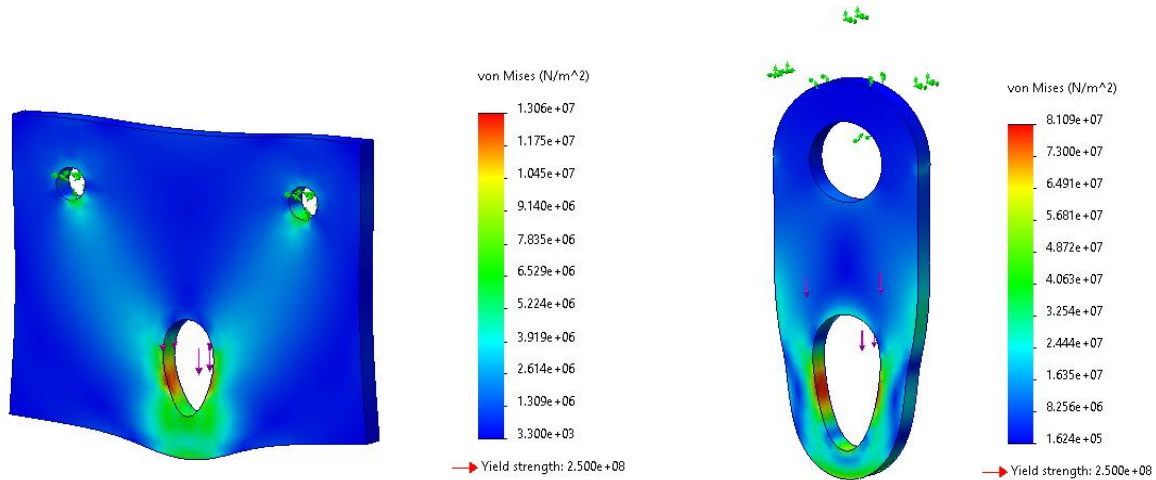


Figure 18: Chain hoist mount tab and plate FEA.

Linear Rails and Hard Stops

Due to the single point that the chain hoist pulls at, the entire carriage requires alignment not only during height adjustment, but also during the drop sequences to keep packages at their required orientations before drops. To do this, linear rails are mounted to the steel tube structure, paired with their subsequent linear bearings on the carriage to keep the carriage aligned during machine operation. These off-the-shelf linear rails have a lower tolerance requirement than other, higher precision linear rails. Due to the large assembly and tolerances the team was able to hold during manufacturing, this lower tolerance was beneficial for the machine to reduce binding. These linear rails are placed at each corner of the structure, which map to each corner of the carriage.

At the bottom of these linear rails are steel hard stops which allow the carriage to sit freely without tension in the chain hoist. These are included not only for ease of assembly, but also as a mechanical failsafe to ensure that the carriage/drop mechanism never hits the ground if there is an error with the height adjustment system or drop sequence. These hard stops physically impact the carriage edges before it moves low enough to touch the impact surface.

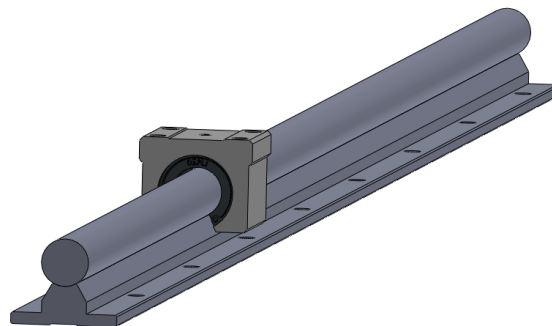


Figure 19: Linear rail and bearing system.

Impact Surface

The primary design driver of the impact surface was ASTM 5.1.5.1, which dictates the drop surface to be concrete or steel, and if steel, it must be at least 0.5" thick. Due to box geometry and the overall footprint of the packages that the machine will drop, a 3' x 5' overall machine footprint was used to accommodate both the main structure, as well as the allocated drop surface area. In the impact surface, there are ten 1/2" tapped holes for connecting the column structure to the base, with six additional 3/8" tapped holes around the edges for lifting eyes to make machine relocation easier. Calculations for the center of gravity and required forces to tip the machine can be found in Appendix A.

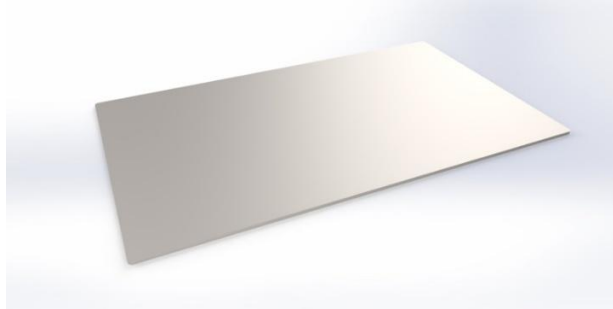


Figure 20: Impact surface.

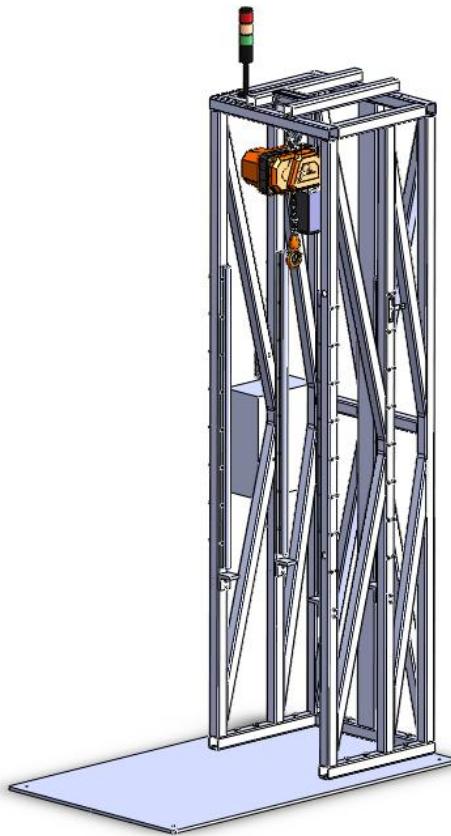


Figure 21: Main structure, chain hoist, and linear rail system.

Column Structure

The structure is designed to support the carriage and absorb the impact from drop sequence. Figure 21 illustrates the column structure design, which includes four linear rails and bearings used to support the carriage and twenty-three individual members. The rails are mounted on two members on each side in a square shape to resist torques that result from a package being placed on the machine, as well as the forces created by the leaf drop sequence. The primary structure is constructed out of welded steel square tubes, with nineteen 2" x 2" x 0.120" members and one 2" x 4" x 0.120" member. Square tube steel was selected for its high strength value and affordability. On top of the main structure, three 2" x 2" x 0.25" members were added to support the chain hoist. The static deflection analysis for a 1200 lbf point force, seen below in Figure 22 on the left, indicates that structure will deflect a maximum of 0.005" under a peak load of 1200 lbs. The results of the static stress analysis, shown in Figure 22 on the right, show that the structure has a minimum factor of safety of 3.

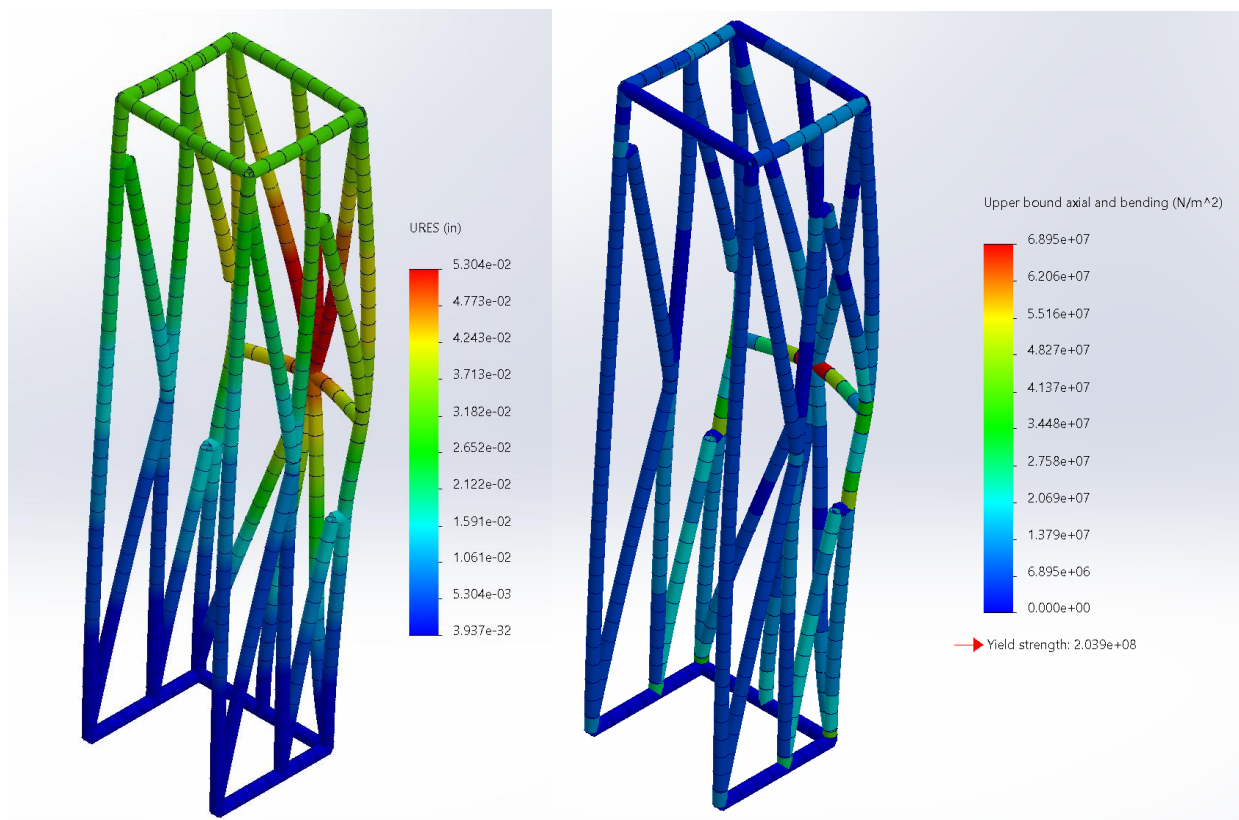


Figure 22: Finite element analysis of structure. Static loading deflection analysis (left). Static stress analysis (right). A point force of 1200lb was applied at the center of the rear member.

Iterations

The first design for the height adjustment system used a lead screw with bearings on each end to mount it to the structure. The lead screw used for this height adjustment system was a Single-start 72" 1"-5 Precision Acme Lead Screw. A single start lead screw was selected for its self-locking property that only allows the nut to move when the lead screw is being turned.

The upper end of the lead screw was connected to a shaft coupler, directly driven by a motor on top of the machine. At the bottom of the lead screw, there was a radial bearing and thrust bearing to handle any unintentional lateral or axial loads.



Figure 23: Old lead screw and bearing mount.

Following the manufacturing review, it was decided that a thorough redesign of the structure and height adjustment system was needed for success of the project. It was noted that the large lead screw used for adjustment was a large risk as it needed to be kept strictly aligned to work properly. Due to the length of the lead screw, the uncertainty in its behavior under load, and the complexity of ensuring its alignment, it was decided to pivot to a chain hoist system. The chain hoist was selected for the lessened alignment constraints, higher reliability, and replaceability.

Electrical System & Human Machine Interface

Electrical Design Overview

The electrical system that powers and controls the machine is designed to be simple, efficient, robust, and easy to service. All major components are housed in an enclosed electrical box mounted on the side of the machine for easy access and maintenance. Each component is clearly labeled and organized for a clean, professional layout.

The system is designed around 120VAC input voltage and converts this power to 12VDC and 24VDC for the main system and the safety circuit respectively. The system also features redundant safety systems to ensure proper operation. This system was designed with poka-yokes in mind which includes E-stops, limit switches, and interlocks that monitor motion and limit positions to ensure safe operation under all conditions.

System Architecture and Components

The system operates from a 120VAC main input, which is converted to 24V and 12V through two dedicated power supplies. The 24V line powers the dedicated safety circuit line for two contactors triggered by the safety relay. The 12V line powers all other electrical components, including the status light and main PCB which distributes power to all other components. An Arduino Nano Every serves as the central controller, processing feedback from the controller and limit switches and commanding the pneumatics, chain hoist, and tower light.

Safety devices include both soft and hard emergency stops. The hard E-stop located on the front of the electrical box is wired to force immediate system shutdown, where all machine components will instantaneously depower and the pneumatics will depressurize. The soft stop on the controller triggers a controlled safe state through the Arduino's logic. In this state, machine components will still be powered but motion is restricted.

The system draws a maximum of approximately 500 W when the chain hoist is active at peak load. At 120 VAC, this corresponds to a current draw of less than 5 A, which is within the capacity provided by Micro Motion.

Table 2 below lists all primary electrical components used in the system along with their functions and design rationale.

Table 2: Primary electrical components.

Item	Function	Rationale
Pro Winch PWBDH660RU	Chain hoist for raising and lowering the carriage	Chosen for its wireless capabilities, making it easier to integrate with the system.
MAKERELE 12V MDR DIN Rail Power Supply 20W, AC 100V-240V to DC 12V 1.67A Power Supply Switch Converter	Provides 12V DC power to system	Supplies power to all electronic components except safety circuit and chain hoist. Chosen for its din rail mount and cost-effectiveness.

Item	Function	Rationale
MAKERELE 15W 24V DR AC to DC Slim DIN Rail Power Supply, AC 100V-240V to DC 24V Power Supply Switch Converter	Provides 24V DC power to system	Powers contactors for the safety circuit. Chosen for its compatibility with din rail, cost-effectiveness, and ease of use.
Arduino Nano Every	Primary microcontroller	Compact, cost-effective, and offers enhanced processing performance. Meets all I/O requirements and easily integrates on the system PCB.
ME-8108 Adjustable Roller Lever Arm Limit Switch	Detects upper and lower travel limits of the carriage	Prevents mechanical overtravel and potential damage. Inexpensive and allow for overtravel in the event the carriage does not stop immediately when triggered.
SS-10GL2 Limit Switch	Detects upper and lower travel limits of the leaf	Prevents mechanical overtravel and potential damage. Inexpensive and compact switches meet the safety and positional feedback needs of the drop mechanism system.
HB-K20 Waterproof IP65 22mm 20A Emergency Stop Mushroom Head Push Button Switch 2NO 2NC	Emergency stop for system shutdown	Critical safety device allowing immediate power cutoff. Chosen for high visibility, accessibility, and ease of operation.
Adafruit Tower Light w/ Buzzer	Provides visual and audible status indication	Cost-effective solution for communicating system states to nearby personnel. Features three lights and a buzzer.
Shneider Electric XPSUAF33AP	Dedicated safety relay	Chosen for its robust safety operation. In the current configuration it will force system shutdown if there is loose connections, broken wires, or a short.
CN0295D LCD screen w/ I2C	Displays carriage height, status, and errors on the controller	Simple and effective interface for user feedback. I ² C communication minimizes wiring complexity and allows easy integration with the microcontroller.

The overall electrical schematic for the system can be seen in Figure 24. The main power switch is wired in series with the 120VAC input. Two glass fuses protect the Arduino and PCB along with a 10A circuit breaker that protects the chain hoist. Each fuse is easily replaceable for maintenance.

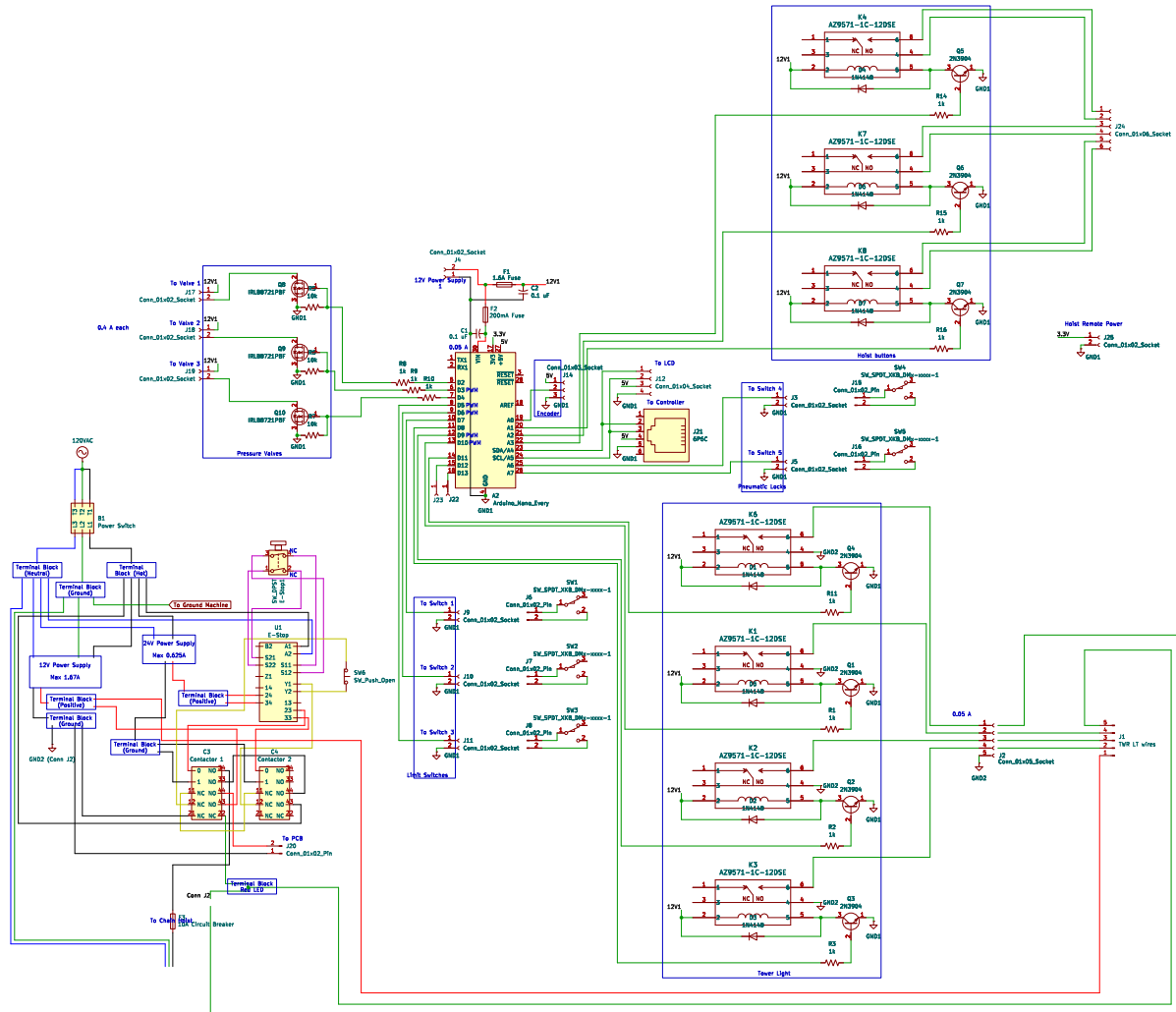


Figure 24: Overall power and safety circuit seen to the left. Main PCB wiring to the right.

Power Supply Specifications

The power supplies used are both MAKERELE products and have an output power of ~20 W. They can accept an input voltage range of 100 – 240 VAC. Both power supplies have short circuit protection, over-voltage protection, overload protection, earth leakage protection, overcurrent protection, and overheat protection.

Arduino Nano Every Specifications

The Arduino Nano Every uses an ATmega4809 microcontroller and has a clock speed of 20MHz. The board operates on 5V and can take an input voltage of up to 12V. The board was chosen for its 20 I/O ports which can optionally be configured to be digital pins. It is also one of the faster boards of the Arduino family, is affordable, and is easily PCB mountable.

Electrical Box

The electrical box is an IP67-rated plastic enclosure junction box similar to the one shown in Figure 25. The box is secured to the machine's frame using three mounting brackets. A lock is provided to prevent unauthorized access to the box.



Figure 25: OTS junction box for housing main electrical components.

As illustrated in Figure 26, the internal junction box layout prioritizes cable routing and ease of maintenance. Power supplies are positioned near the bottom of the box close to the power switch. Terminal blocks distribute power to connected subsystems, while the main PCB and controller are located at the top of the enclosure. All external connections exit through a single side-mounted port. The E-stop and power switch are mounted externally for clear visibility and accessibility. An LCD screen is mounted to the front of the box and displays instructions and error messages in case of a fault.

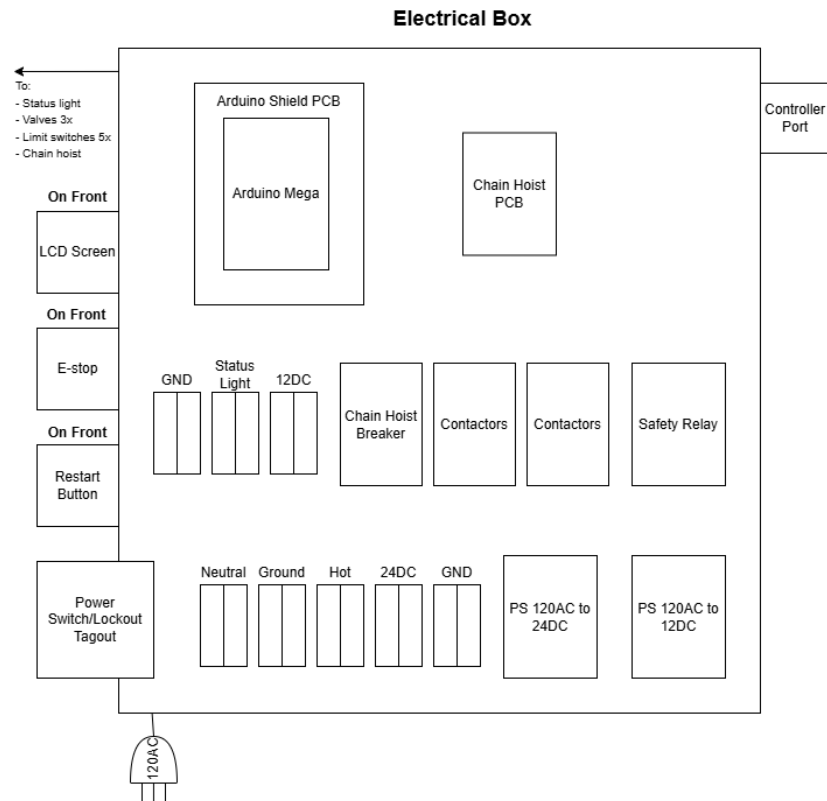


Figure 26: Internal junction box layout.

Arduino PCB

A 3D render of the main control PCB is shown in Figure 27. The board incorporates BJT switches for controlling major components. Communication with the handheld controller is handled via an RJ11 cable, chosen for fast communication and minimal wiring. To address vibration concerns from machine operation, JST-XH locking connectors are used to prevent cable disconnection or loosening over time.

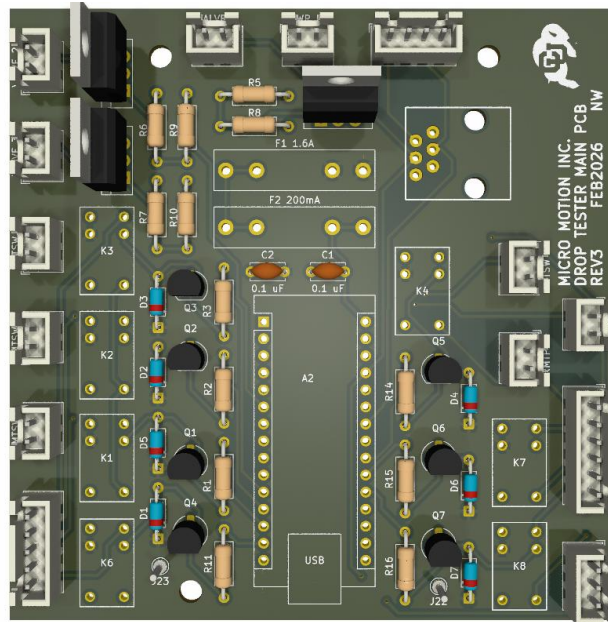


Figure 27: Main PCB.

Cable Management

Due to the number of moving parts on the machine, cable management is very important. All of the wires around the machine are routed on the outside of the frame and are protected with a sleeve to protect them from damage.

The wires routed to the pneumatics and limit switches on the carriage must move up and down by 36" without getting caught in the linear rail system. To keep the wires safe, they are routed through a cable-carrier chain. One side of the cable-carrier chain is mounted to the frame at the top of the carriage travel, and the other side is mounted to the carriage. This allows the wires to travel up and down with the carriage while keeping them on a known path away from the linear rails. The cable-carrier chain is pictured below in Figure 28.



Figure 28: Cable-carrier chain.

The controller is stored in a holder on the machine but must be able to be moved several feet away from the machine for the operator to use the machine safely. To keep the wires out of the moving pieces of the machine, a curly 4 strand telephone cable is used. This cable retracts to 1.2 ft and extends up to 10 ft allowing the user to step a safe distance away from the machine while using the controller.



Figure 29: Curly telephone wire for the controller with RJ11 connectors.

Human Machine Interface

The human machine interface is how the user interacts with the machine and operates it. During normal operation, the user interacts with the electrical box and the controller. The electrical box is mounted on the left side of the machine and has a power switch, E-stop, restart button, and LCD screen on the outside of the box. The main electrical components are mounted inside of the box and should not need to be accessed during normal operation. The controller is stored in a holder located above the electrical box and has a stretchable cord to allow the user to step away from the machine during operation. Both the electrical box and controller holder are pictured in green in Figure 30 below.

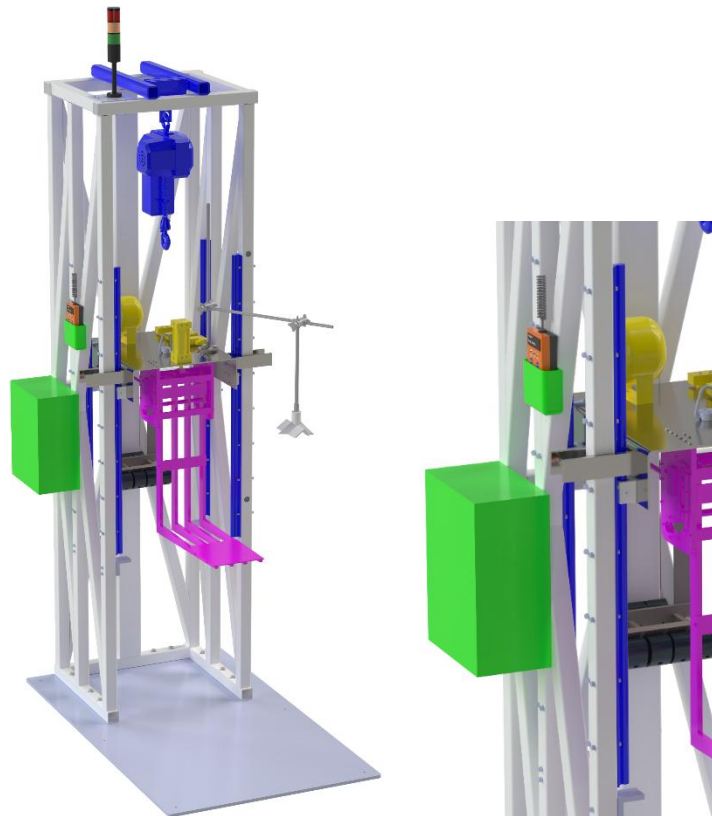


Figure 30: Electrical box and controller holder placement.

The other way users interact with the machine is the actual placement and securement of the package. The user will place the package on the drop leaf and then use the angle fixation claw to hold the package in the desired orientation.

To indicate the status of the machine to the user and anyone else around it, there is a status light located on top of the machine. Figure 31 below outlines the meaning of each color on the status light.



Solid Color	
Red	E-Stopped
Yellow	Soft Stopped / Initialization
Green	Height Adjustment Enabled
Flashing Color	
Red	Error / Fault
Yellow	Armed / Height Adjust Moving
Green	Starting Machine
Buzzer	
2 Sec	Before Drop / Reset

Figure 31: Status column light indicator colors.

Controller Design and PCB

The controller is the main way the user operates the machine. The controller has eight buttons which allow the user to control the machine's movements. These buttons are: Stop, Start, Up, Down, Arm Drop, Drop, Arm Reset, and Reset. The machine indicates what state it is in through the status light and LCD screen. The user manual and LCD screen provide instructions to the user.



Figure 32: Machine controller.

The controller has a case that houses and protects the electrical components inside of the controller. Inside of the case, there is a custom PCB with eight buttons, one MCP23008-E/P I/O expander chip, and an RJ11 connector for the wire going back to the electrical box. This quick disconnect RJ11 connector allows the controller to be disconnected from the machine for ease of transportation and repair. The I/O expander chip communicates over I2C serial communication

with the Arduino in the electrical box. The I/O expander chip has eight bidirectional pins with internal pull-up resistors. Custom button covers were created out of silicone to provide a good user experience and to label the buttons. The controller PCB design is shown below in Figure 33. The wiring schematic for the controller PCB is shown below in Figure 34. CAD models of the button covers and mold are shown in Figure 35.

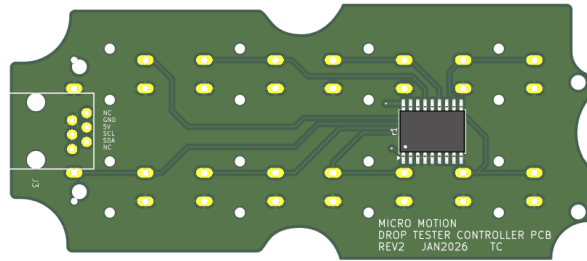


Figure 33: Controller PCB.

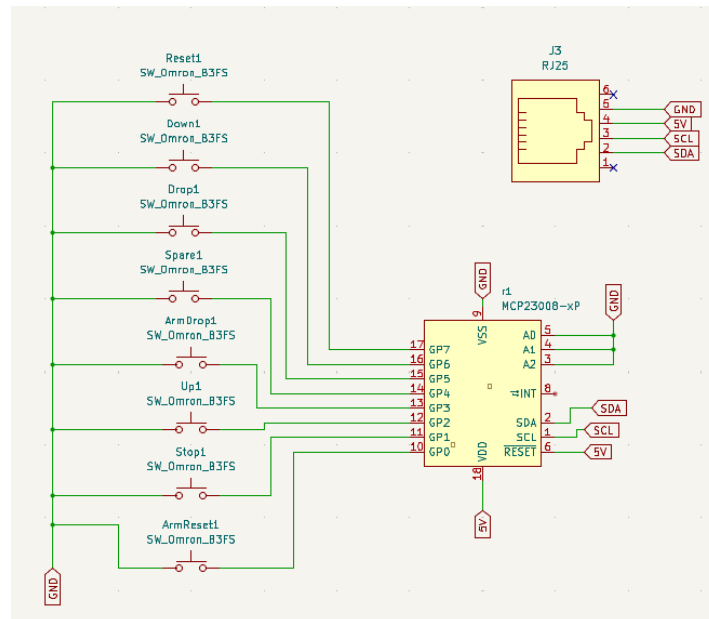


Figure 34: Controller wiring schematic.

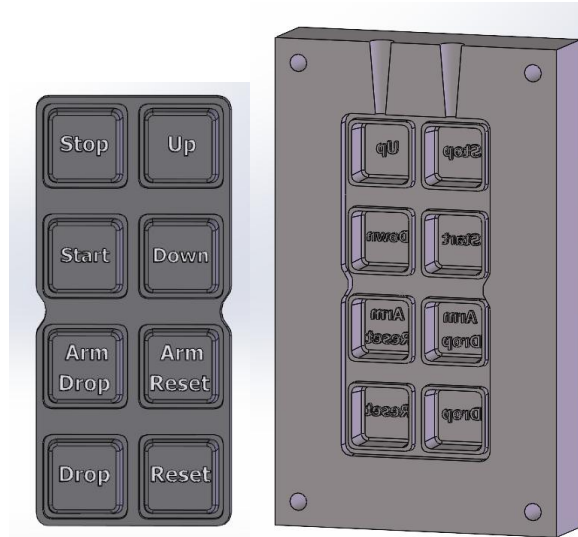


Figure 35: Controller button covers and mold.

Manual Height Readout

The drop height of the leaf is controlled by using the Up and Down buttons on the controller. Pressing either button triggers the chain hoist, which lifts or lowers the carriage. The user reads the current drop height using a ruler and height indicator located on the side of the machine. An image of the height indicator is shown below in Figure 36.



Figure 36: Manual height readout.

Software

The software is written in Arduino IDE and is run on the Arduino in the electrical box. This Arduino controls the machine by reading the sensors and buttons and controlling the chain hoist and pneumatics. The software controls the machine in a safe way by going through multiple safety checks before initiating any movement. This includes checking the sensors to make sure that it is safe to move the carriage up or down and only allowing certain movement in each state to avoid unsafe behavior.

The software is structured using a state-based machine framework. The state diagram shown below in Figure 37 represents the behavior of the machine, showing all possible states it can be in and what causes the transitions between the states.

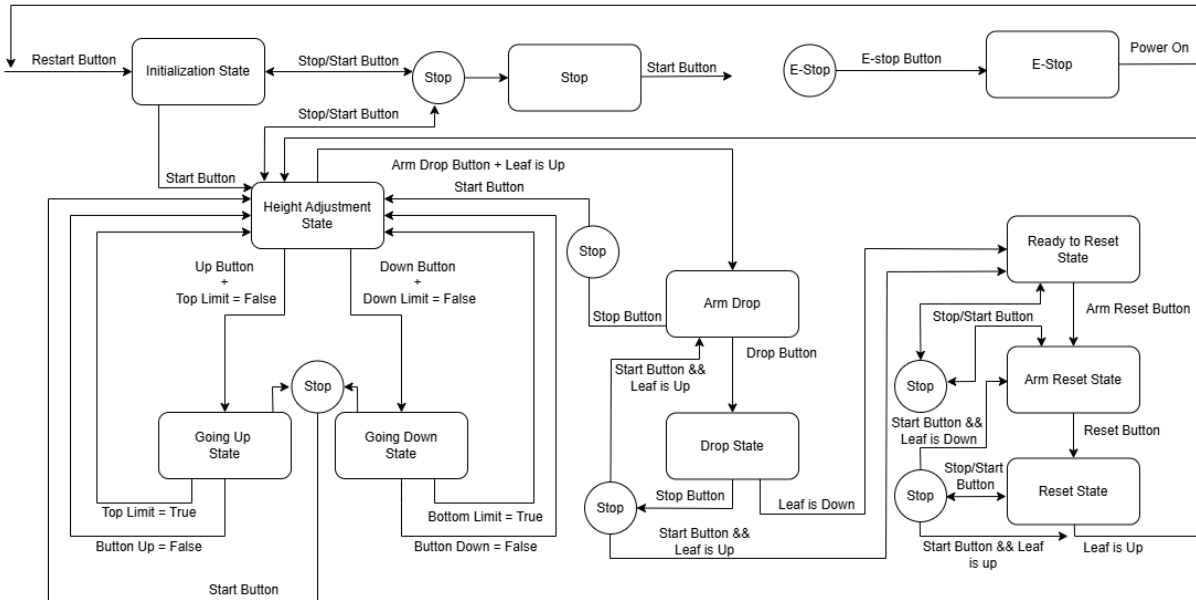


Figure 37: State diagram outlining the operation of the machine.

Once the power switch has been turned on and the Restart button on the electrical box has been pushed, the machine goes into an initialization state. During initialization, the Arduino boots up and establishes connections with the controller and the sensors. Once complete, the user is prompted to hold the Start button on the controller for two seconds, after which the machine transitions into the height adjustment state. This state allows the user to adjust the height up and down by using the Up and Down buttons on the controller. The height is limited by limit switches placed at 12" and 48" on the structure.

From the height adjust state, the user can press the Arm Drop button which will put the machine into a state where it prepares for the drop by readying the pneumatics. At this point, it disables height adjustment until the end of the drop and reset sequences. Next, it moves into the drop state and drops the package when the user presses the Drop button. After the package has been dropped, the machine automatically goes into a ready-to-reset state, where the user must reset the leaf before doing anything else. This is done by pressing the Arm Reset button followed by the Reset button, which brings the leaf up to the top position. Once the leaf is all the way up, the machine returns to the height adjustment state.

At any point in the operation of the machine, if the Stop button on the controller is pressed, the software will stop the movement of the machine. To get out of the stop state, the Start button must be pressed, which causes the machine to go to the last safe state. For example, if the Stop button is pressed in the drop state and the leaf has not fallen yet, the Start button will cause the machine to go back to the arm drop state. If the leaf has fallen, the Start button will cause the machine to go to the arm reset state. This behavior makes sure that no sudden movements happen when returning from the stop state. It is also useful for aborting a drop if the user decides they no longer want to drop a package. If the E-stop is pressed at any point, the power to the entire machine will be cut. To get out of the E-stop state, the E-stop must be physically reset and the Restart button pushed. This resets the machine back to the initialization state.

Iterations

Throughout the project, there were many iterations and redesigns of the electrical system. The main PCB underwent three iterations. Version 1 of the PCB was a smaller footprint originally designed for the system before the redesign that occurred over winter break. Due to the drastic change in system architecture, a new PCB design was needed. Version 2 is visually identical to the final PCB, but it was discovered that a catastrophic error in the power input traces rendered the PCB completely inoperable. PCB version 3 was the final design for the project, implementing fixes to the power input and correcting relay orientation.

Late in the design, Micro Motion requested a safety circuit redesign so it would fully comply with their safety standards. This included adding a safety relay, two contactors, a reset button, and changing the E-stop. The safety circuit was designed around the Schneider Electric safety relay. This relay is a robust and trusted option chosen with the help of Micro Motion. The relay is directly powered with 120VAC input and communicates with 2 contactors, the E-stop, and a reset button via 24VDC. The contactors are responsible for mechanically switching when the safety relay switches states. Under normal operation, the chain hoist, PCB and pneumatics will receive power, but if the safety relay detects a fault, or is manually triggered, the contactors will flip, switching the power to the alternate path. This structure was used so that the tower light located on the top of the machine could be hardwired. Normally, the tower light is triggered by the software running on the Arduino, but red light is hardwired so that if there is a fault the red light will illuminate, despite the Arduino and main board being turned off.

The controller was initially designed to have a 3D printed case and an LCD screen on the controller. However, the chain hoist arrived with a nice case that was not needed, so this case was used instead. A PCB was designed to fit inside of it, and the case was modified to allow for a wired connection to the electrical box. This meant that the LCD screen had to be moved to the electrical box instead of being on the controller. Custom silicone button covers with labels were also designed to make the controller easy to use. These button covers took multiple iterations of molds and materials. The first attempt used 3D printed resin for the mold and platinum-cured silicone for the buttons. The silicone reacted with the off-gases from the 3D printed resin and never cured. The next iteration used tin-cured silicone, which cured well but was very viscous, causing filling issues and trapped air bubbles. The final version was made using the tin-cured silicone and pouring it into the front of the mold, and then pressing the back of the mold on. The letters were painted on using white dye in platinum-cured silicone. This allowed the paint to cure to the buttons and be flexible, so it did not crack when the buttons were pressed.

Initially the design had a linear encoder for a digital height readout. This allowed full control over the height with the control system and allowed for 2" height increments. However, the violent movement of the carriage proved to be too much for the linear encoder which was placed on the frame and had a wiper that triggered it from the carriage. The encoder was damaged from this movement, and a manual height readout system was implemented instead. This system has a ruler placed on the side of the machine and allows the user to use the Up and Down buttons on the controller until the leaf is at the desired height. This does not specifically only allow 2" height increments but instead allows the user to select any height they choose. This

gives them more flexibility and still meets the tolerance of +1” and -0”. In the future, a distance sensor could be installed on the top of the machine to read the height from the carriage.

Finally, the top and bottom height limit switches went through two iterations. The first design used simple limit switches on spacer blocks which would be triggered by the carriage stabilizers. However, at full speed, the chain hoist did not stop fast enough and destroyed the limit switches. The new design uses adjustable rotary limit switches that are mounted on the frame and trigger off of the carriage. This design works well and keeps the bodies of the limit switches out of the path of the carriage, preventing damage.

Risk Assessment

A Failure Mode and Effects Analysis (FMEA) was done to ensure that all potential risks that can occur on the machine are tested and verified. This verifies that the mechanical, electrical, and pneumatic components work well individually before integrating them with one another. It also helped determine which components and specifications needed more time during testing.

Table 3 and Table 4, below, show the Assembly/Mechanical Failure and Electrical Failure FMEA done for the project.

Table 3: Assembly risk assessment

FAILURE MODE	FAILURE CAUSES	FAILURE EFFECTS	O	D	S	RPN
Track Plate to Carriage Misalignment	Loose bolts and nuts. Incorrect stack-up tolerances.	Critical since all components attached to the Track Plate is important for performing the Drop Cycle	5	7	9	315
Drop Leaf Piston to Drop Cylinder to Carriage Interface Misalignment	Incorrect stack-up tolerances.	This misalignment would not allow the drop cylinder to actuate the drop leaf. Critical because machine fails to meet its functionality requirements.	5	7	6	210
Wiper Bracket to Carriage Interface Misalignment	Loose bolts and nuts. Incorrect stack-up tolerances	Misalignment hinders the Height adjustment system from reading heights. Critical since we need to ensure that the packages are drop at the heights required for it.	5	7	5	175
Linear Bearing to Carriage Interface Misalignment	Loose bolts and nuts. Incorrect stack-up tolerances	Misalignment hinders the Height Adjustment system from moving the carriage smoothly	5	7	5	175
Carriage Stabilizer to Carriage Interface Misalignment	Loose bolts and nuts. Incorrect stack-up tolerances	Critical because the Carriage Stabilizer	5	6	5	150
Damper Assembly to Carriage Interface Misalignment	Loose bolts and nuts. Hole misalignment between both interfaces	Damper Assembly fails to hit the back of the Drop Leaf. Failure can cause the Drop Leaf to push the back structure members that can cause permanent damage on both components, leading to complete component failure.	5	7	8	280
Drop Leaf Axle to Drop Leaf Interface Misalignment	Press fit tolerances are off. Snap Rings fail to fit and align with both interfaces	Drop Leaf will not fit on to the Track Plates and Drop Cylinder. Critical since the Drop Leaf is an integral part of the machine and failure to keep this aligned will cause the whole machine to fail all of its required functionalities.	3	5	6	90
Linear Guide to Structure Member Interface Misalignment	Loose bolts and nuts. Incorrect stack-up tolerances	Carriage will not be raised and lowered smoothly. Failure leads to complications on the mechanical side of the height adjustment system that can cause machine to fail it height adjustment requirement	3	8	5	120

FAILURE MODE	FAILURE CAUSES	FAILURE EFFECTS	O	D	S	RPN
Carriage Hard Stops to Structure Member Interface Misalignment	Loose bolts and nuts.	Carriage is not be fully supported when it rests in the Carriage Hard Stops. Failure can break the Drop Mechanism assembly and can cause the machine to break	3	8	4	96
Chain Hoist Plates to Structure Member Interface Misalignment	Loose bolts and nuts.	Chain Hoist is misaligned with the top of the carriage. Failure causes Chain Hoist to lift the carriage in a position that doesn't support its center of gravity. This will cause the machine to lose its balance during the Drop Cycle and can lead to the machine breaking itself.	3	7	10	210
Chain Hoist Plates to Clevis Pin Interface Misalignment	Loose Clevis Pin.	Clevis Pin falling out of the Chain Hoist tab can cause the Chain Hoist and all Mechanical Height Adjustment components to fall on to the carriage. This will cause severe harm on the operator and its surroundings.	2	7	10	140
Locking Cylinder Assembly to Track Plate Interface Misalignment	Loose bolts and nuts.	Critical since the Locking Cylinder Assembly is responsible for pushing the Leaf Locking Tab in place and pulling it back before the Drop Cycle. Misalignment causes Drop Leaf to smoothly drop during the cycle as well as locking it in place while its at its resting position.	3	5	9	135
Track Guides to Track Plate Interface Misalignment	Loose bolts and nuts. Incorrect Stack-up Tolerances	Misalignment cause the Drop Leaf to shift constantly. Critical because it needs to stay still at all times or else the machine fails to meet all of its machine functionality.	2	8	8	128
Limit Switch Spacer to Track Plate Interface Misalignment	Loose bolts and nuts. Incorrect Stack-up Tolerances	Misalignment will hinder the limit switches from performing their corresponding functions. Critical because this lowers the machines safety reliability.	3	6	7	126

Table 4: Electronics failure risk assessment

FAILURE MODE	FAILURE CAUSES	FAILURE EFFECTS	O	D	S	RPN
Emergency Stop on Electrical Box fails to cut power to the machine when pressed	Emergency Stop Button is disconnected. Code did not initialize properly.	Severe harm on the operator and its surroundings, can cause death. Machine breaks leading to its complete failure.	3	5	10	150
Stop Button Functionality Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Severe harm on the operator and its surroundings, can cause death. Machine breaks leading to its complete failure.	3	5	10	150
Start Button on Controller/ Initialization Mode Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Controller fails to turn on the machine. Critical since this leads to the machine's complete failure	3	5	8	120
Reset Button on Electrical Box Failure	Reset button is disconnected. Code did not initialize properly.	Fails to re-initialize the machine after pressing the Emergency Stop. Can cause the miscommunication between the controller and the operator, which can cause severe harm on them.	3	5	10	150
Up Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Chain Hoist is disconnected. Code did not initialize properly.	Controller fails to alert the Chain Hoist to raise the Drop Leaf. Critical since machine fails to meet its required height control requirement.	3	5	9	135
Down Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Chain Hoist is disconnected Code did not initialize properly.	Controller fails to alert the Chain Hoist to lower the Drop Leaf. Critical since machine fails to meet its required height control requirement.	3	5	9	135
Arm Reset Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Controller fails to alert the Drop and Locking Cylinders to depressurize. Failure will cause Drop Leaf to stay stuck in its current position	3	5	8	120
Reset Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Controller fails to alert the Drop and Locking Cylinders to depressurize. Critical since Drop Leaf can't come back up to its resting position	3	5	10	150
Arm Drop Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Controller fails to alert the Drop and Locking Cylinders to pressurize. Failure hinders the Drop Cylinder to actuate the Drop Leaf and hinders the Locking Cylinder to retract before the Drop Cycle	3	5	8	120
Drop Button on Controller Functionality Failure	Controller is disconnected from the electrical box. Code did not initialize properly.	Controller fails to alert the Drop and Locking Cylinders to pressurize. Critical because the Drop Leaf will not actuate, failing its Drop Cycle requirement	3	5	10	150
Wiper to Linear Encoder Interface Failure	Wiper adding to much pressure on the Linear Encoder physically cuts it in half	Linear Encoder will fail if wiper cuts it in half, leading to the component's complete destruction. Critical since the machine fails to have an automated height readout	3	4	7	84

FAILURE MODE	FAILURE CAUSES	FAILURE EFFECTS	O	D	S	RPN
Height Adjustment Limit Switches Failure	Limit Switch is disconnected. Limit Switch fails to touch the carriage	Failing to read height causes the carriage to go either past 48" or 12". Critical since this can cause the Drop Mechanism and/or the whole machine to break	3	6	9	162
Drop Cylinder Limit Switch Failure	Limit Switch is disconnected	Fails to read if the Drop Cylinder has returned to its resting position. Critical since it indicates that the Drop Leaf failed return up.	3	6	9	162
Locking Cylinder Limit Switch Failure	Limit Switch is disconnected	Fails to read if the Locking Cylinders have retracted to its resting position. Critical since it indicates that the Drop Leaf failed return up.	3	6	9	162
Status Light flashes the incorrect color for each state	Initialization failed. Re-initialization after resetting machine fails.	Critical because this affects the reliability of the machine's safety response	2	6	9	108

Test Results

To ensure the package drop tester meets the requirements specified by Micro Motion and verify the functionality of the final machine, Team 23 developed a series of test plans. These tests were based on the Project Requirements table (Table 1), the need to verify the functionality of critical components, and assess potential safety concerns. The tests were split into the following categories: Safety, Package Requirements, Electronics and Machine Functionality.

Safety & Machine Functionality

One of the most important requirements of this project is the safety of users and ease of operation. These fundamentals are the basis for many design choices and were focused on in testing. Pressure tests were conducted for the pneumatic system, with the use of a pressure regulator to adjust the dynamics of the drop leaf. Additionally, the functionality of safety components such as the limit switches that restrict carriage motion, e-stop, and status light were verified. Based on the results of these tests, the machine works as Micro Motion specified and intended.

Results

As a result of testing the dynamics of the drop leaf, it was found that as it impacted the dampers, it bounced and impacted packages on their way down. To mitigate this bounce from the drop leaf, less stiff dampers were implemented. It was also found that the drop leaf retracted back into place faster than expected and caused a potential safety hazard. This facilitated more design changes, adding a throttling valve to adjust the pressure of the drop leaf retracting. When testing carriage motion along the linear rails, there was significant binding. This resulted in adding shims to offset the warped frame, allowing the carriage to move along the linear rails more smoothly. The limit switches used to restrict the carriage from moving off the linear rails were tested at full speed. The original limit switches used could not stop the carriage without getting damaged, resulting in another design change to more robust limit switches. The final limit switches can safely stop the carriage at full speed, restricting the carriage from moving off the linear rails. Finally, it was found that the status light indicated each machine state and machine motion correctly.

Package Verification Test

Objective

The goal of the package verification test was to ensure that the drop machine properly supported packages ranging from the minimum package size to the maximum package size, and a weight up to 90 lbs. This test demonstrated the machine's ability to support these boxes in 10 different orientations according to Micro Motion's drop standard, which follows ISTA 1C/1E Standards. This test simulated Micro Motion's testing procedure to ensure proper machine functionality and package accommodation. The package verification test included dropping packages of sizes specified in Table 5 for 10 drop tests, one for each box orientation specified by ISTA 1C/1E, for a total of 60 drop tests.

Table 5: Package dimensions and weight ranges.

Length (in)	Width (in)	Height (in)	Approx Weight Range (lb)
35.25	29.88	14.63	20-90
39.5	43.25	15.5	25-95
37.5	29.5	14.25	10-50
33.56	25.63	13.63	25-80
12.69	9.69	13.31	10-40
19.31	13.63	12.81	15-40

Measurements & Procedures

Package drop measurements were entered into an excel spreadsheet labeled with the package dimensions and weight. These measurements included box orientation, drop leaf deflection, weight, if the box has been correctly supported, reload time, if the individual drop test passed or failed, and additional notes on the drop test. A drop test was marked a failure if the drop leaf deflection was greater than 2° (ASTM D5267-19), or if the package could not be properly supported in any of the box orientations. Additionally, the reload time had to be less than 15 seconds. Each drop test was recorded using a camera and tripod to ensure package interference does not occur and no information is lost.

The following steps will be performed for each drop test:

1. Record package size and weight on testing table
2. Record the end of the drop leaf (top edge) height with no package (to check for deflection)
3. Record orientation of box on testing table
4. Support box with gooseneck assembly (record T/F on table)
5. Record the end of the drop leaf base (top edge) height with package
6. Drop package
7. Record any package interference
8. Reload (record reload time)

Results

Table 6: Drop test measurements for smallest package size.

Test Trial #	Tester Name	Date	Height Readout	Measured Height	Face/Orientation	Reload Time	Weight	Supported	Pass/Fail	Comments
1	Francine	12-Mar	-	12"	Corner 2-3-5	12.08	2lbs	☑	F	Drop leaf impacted package after bouncing off dampers
2	Francine	12-Mar	-	12"	Edge 2-5	15	2lbs	☑	F	Drop leaf impacted package after bouncing off dampers
3	Francine	12-Mar	-	12"	Edge 2-3	18	2lbs	☑	F	Drop leaf impacted package after bouncing off dampers
4	Isaac & Teagan	13-Apr	42"/14"	-	Bottom Face	10.8	empty	☑	P	Passed, Height readout was not working at the time so Height measurement was not taken
5	Isaac & Teagan	13-Apr	42"/14" (inaccurate)	-	Bottom Face	11	empty	☑	P	Height readout not working
6	Isaac & Teagan	13-Apr	42"/14" (inaccurate)	-	Bottom Face	12	empty	☑	P	Height readout not working
7	Expo Demonstrat	17-Apr	36"	36 1/8"	Corner 2-3-5	11	empty	☑	P	3-d printed indicator was temporarily glued on for expo
8	Expo Demonstrat	17-Apr	36"	36 1/8"	Corner 2-3-5	10.9	empty	☑	P	
9	Expo Demonstrat	17-Apr	36"	36 1/8"	Corner 2-3-5	11	empty	☑	P	
10	Expo Demonstrat	17-Apr	36"	36 1/8"	Corner 2-3-5	12.1	empty	☑	P	
7	User Testing	21-Apr	36"	36 1/4"	Corner 2-3-5	11.2	5lbs	☑	P	Switched to manual readout, updated gooseneck design
8	User Testing	21-Apr	30"	30 1/8"	Corner 2-3-5	10.9	5lbs	☑	P	
9	User Testing	21-Apr	40"	40 1/8"	Corner 2-3-5	11	5lbs	☑	P	
10	User Testing	21-Apr	35"	35 1/8"	Corner 2-3-5	12	5lbs	☑	P	Note: For user testing I filled the box with our fastener bags

Table 7: Drop test measurements for largest package size.

Test Trial #	Tester Name	Date	Height Readout	Measured Height	Face/Orientation	Reload Time	Weight	Supported	Pass/Fail	Comments
1	Isaac & Tegan	April 14th	12"	12.125"	Bottom Face	11.14s	90lbs	☑	P	Package was not hit on it's way down but it was impacted after drop was complete
2	Isaac & Tegan	April 14th	12"	12.125"	Bottom Face	11.41s	90lbs	☑	P	Package was not hit on it's way down but it was impacted after drop was complete
3	Isaac & Tegan	April 14th	12"	12.125	Bottom Face	11.06s	90lbs	☑	P	Package was not hit on it's way down but it was impacted after drop was complete
4	Isaac & Tegan	April 14th	12"	12.125	Bottom Face	12s	90lbs	☑	P	Note Deflection measured 1 degree for weight of 90lbs for all tests on April 14th

Through this testing, issues were found with key components, such as the dampers, binding on the carriage, and limit switches not engaging. As a result, the team focused on correcting these design errors and had less time to drop as many packages as was initially planned. However, tests were able to be completed on the largest package and smallest package sizes. The machine was found to have a reload time of approximately 12 seconds and 1° of deflection when dropping a 90lb package, satisfying the reload time and 2° requirements.

Software and Electronics Test

Objective

The objective of this test was to ensure proper communication between the Arduino and PCB, as well as verifying the software functioned as intended and safety features operated as expected. Additionally, this test aimed to confirm the two button sequences for resetting and dropping operations of the package drop tester. Data collected was used to verify the software function and adjust software logic as needed.

Procedure

Software testing required the setup of all electronic components and wiring each component to create a mock system with the pneumatics. This allowed for testing software and electrical components before integrating with the final assembly. To test the dropping sequence and software, buttons on the controller were pressed one after another, and the resulting actions were recorded to determine if the system made the intended action. This data was stored in a truth table as seen in Table 8. In this table, once a button is pressed, it is marked as true. If a button has not been pressed, it is marked as false. The expected outcome is marked as either true or false, depending on the response of the machine. This button testing was done for a list of button combinations to verify the software logic.

Table 8: Software logic test chart.

Trial #	Up Button	Down Button	Arm Drop Button	Drop Button	Arm Reset Button	Arm Button	Emergency Stop	Expected Outcome	Comments
1	T	T	F	F	F	F	T	T	
2	F	F	T	T	F	F	T	T	
3	T	F	F	T	F	F	F	F	
4	T	T	F	T	F	F	F	T	
5	T	T	T	T	T	F	F	T	
6	F	F	T	T	T	T	T	T	

Results

As a result of testing, several bugs in the software were found that were then fixed. After rounds of further testing, the software logic was confirmed to not have further issues. Some of the results of the testing are seen in Table 9; however, this is not an exhaustive list of operations that were tested. Additional edge case testing was conducted to develop error messages for the LCD screen on the electrical box. The safety circuits, limit switches, and other safety features were found to operate correctly.

Table 9: Software logic tests.

Individual Testing	Trial #	Up Button	Down Button	Arm Drop Button	Drop Button	Arm Reset Button	Reset	Emergency Stop	Start	Stop	Expected Outcome	Expected Outcome Description	Pneumatic operation	Comments	Date
Teagan	1	F	F	T1	T2	T3	T4	F	F	F	T	Drop mechanism performs a drop. Drop leaf drops and resets to starting position.	Pneumatic locking tabs disengage after arm drop, drop cylinder forces drop leaf down after drop button pressed. Release piston pressure after arm reset. Drop leaf retracts & locking tabs engage after reset is	Drop Cycle operation from start to finish. It worked!	4/2/2026
Teagan	2	T2	T3	F	F	F	F	F	T1	F	T	Chain hoist initializes and moves up and down.	NONE	Moved carriage up and down along linear rails successfully	4/2/2026
Isaac	3	T3	T2	T4	F	F	F	F	T1	F	T	Chain hoist initializes and moves down and up, then drop mech is ready for a drop	Locking tab cylinders disengage, drop cylinder ready to fire	Carriage moved correctly, but we got stuck in a endless loop after timing out (just stood idle after pressing arm drop)	4/8/2026
Anselme & Isaac	4	T3	T2	F	F	F	F	F	T1	F	T	Chain hoist initializes and moves down and up	NONE	Moved carriage down and up along linear rails	4/8/2026
Isaac & Teagan	5	T2	F	T3	F	F	T4	F	T1	F	F	Chain hoist initializes and moves up, drop is armed and then reset	Locking tabs disengage, then engage to lock drop leaf	After waiting in the height adjustment state, we timed out and got stuck in a loop. Teagan later fixed the code in the software	4/10/2026
Isaac & Teagan	6	T3	T2	F	T1	F	F	F	F	F	T	Pressing drop should not result in any action, carriage should not move without initializing	NONE	Machine did not drop without arming! Safety	4/10/2026
Isaac & Teagan	7	T3	T2	T1	F	F	T4	F	F	F	T	While in the arm drop state, machine should not move the carriage, and should not reset without arming reset	Locking tabs disengage after hitting arm drop	Carriage didn't move when in the arm drop state	4/10/2026
Isaac & Teagan	8	F	F	T2	F	F	F	T3	T1	F	T	Hitting the emergency stop after arming a drop, powers down the machine and locking tabs should engage	NONE	Machine operated as it should have	4/10/2026

Height Accuracy Test

Objective

The goal of the height accuracy test was to determine the accuracy of the height readout. As shown in the PRS table (Table 1).

this readout must be accurate to +1" and -0". Additionally, it must be able to range from 12"- 48". The height readout was compared to the measured height of the drop, tested for heights 12"- 48" in increments of 2". The pass and fail conditions were based on the +1" and -0" requirement.

Results

As a result of frame warping and binding of the carriage, a digital height readout could not be implemented. Initializing and calibrating the encoder resulted in unreliable height readouts so, after the encoder was damaged, the team opted for a manual height readout. Comparing the indicated height vs. the measured height of the drop leaf platform, it was found that the indicator was 1/8" higher than the drop leaf platform for all positions tested. This was within the +1" and -0" tolerance specified, so the test was deemed satisfactory.

Bill of Materials

The initial team budget from CU was \$2000. This included \$200 for prototyping and \$1800 for the main purchases of materials and components. It was found that the machine would require more than the allotted \$2000, and Micro Motion Inc. provided roughly an extra \$3000 to be allocated towards the project. This projection was determined after doing preliminary quotes and design reviews, where certain ideas had to be scrapped due to certain components being more expensive than the overall budget.

The most expensive single component was the chain hoist, integral in holding up the carriage, as well as providing incremental height adjustment to the machine. Additionally, the steel tubing for the structure and the impact surface for the base plate were also expensive items, contributing to the overall cost of the project. The total for this project was \$5039.37, not including shipping or the cost of labor and free materials the team were able to acquire through the Idea Forge. The team required an additional ~\$350 on the CU procurement card to finish some transactions in the spring semester, for items like paint and a U-Haul to bring the machine to Micro Motion's warehouse. Table 10 below shows a summary of the bill of materials; the full spreadsheet can also be opened with this [link](#) by the Micro Motion representatives (Collin Ruprecht, Mackenzie Lamoureux, or Jack Lattanzi).

Table 10: Bill of materials summary

<u>Micro Motion Package Drop Tester Bill of Materials</u>			
Max Lead Time:	14	Total Cost (TC):	\$5039.37
Most Expensive Part:	\$741.07		
<u>Subassembly Totals:</u>			
Leaf and Carriage:	\$1,712.79		
Main Structure:	\$1,333.16		
Height Adjustment:	\$952.90		
Gooseneck:	\$72.39		
HMI/Electronics:	\$936.18		

Lessons Learned

Design for manufacturability (DFM) was a critical consideration throughout the project. Certain design choices, mainly the use of oddly angled structural members, introduced unnecessary geometric complexity and significantly increased machining time. Additionally, failures such as five broken taps in the 0.5” steel track plates, highlighted the importance of accounting for tooling limitations, material properties, and appropriate thread engagement during the design phase. Greater emphasis on standardization, specifically the use of readily available stock materials such as bar stock rather than custom plate geometries, would have reduced machining effort, improved reliability, and minimized the risk of manufacturing defects. More broadly, incorporating manufacturability constraints earlier in the design process would have enabled more efficient fabrication and reduced iteration during later stages.

From a cost perspective, this project highlighted the importance of evaluating total system cost rather than focusing solely on material expenses. For example, while 80/20 aluminum extrusion would have increased upfront material costs compared to a fully machined frame, it likely would have resulted in substantial reductions in machining time, labor, and overall fabrication complexity. While machining done by the team was free, the Idea Forge machine shop staff frequently emphasized this was not something to be expected in the future. This tradeoff of machining time compared to material costs illustrates the need to balance material selection with manufacturing efficiency during early-stage design decisions, particularly when labor and machining resources are constrained.

The manufacturing and assembly phases also highlighted the importance of tolerance management and alignment considerations. While individual components were designed to nominal dimensions, the accumulation of tolerances and deformation during fabrication, particularly in the welded frame, led to misalignment issues at the system level. This emphasizes the need for tolerance stack-up analysis, as well as the inclusion of adjustment features such as slots or shims to accommodate variation during assembly. Additionally, assembly accessibility and sequencing should be considered during design, as these factors can significantly impact build time and the ability to achieve proper alignment in practice.

Finally, the project demonstrated the risks associated with outsourcing fabrication processes without sufficient specification and oversight. Welding of the frame was delegated to another team with greater TIG welding experience; however, the final assembly exhibited warping that exceeded tolerance limits, leading to later alignment issues. This outcome highlights the importance of clearly defining fabrication requirements, including tolerances, fixturing expectations, and inspection criteria, as well as verifying critical dimensions after outsourced operations. Even when leveraging external expertise, maintaining rigorous communication and quality control is essential to ensure conformance with design specifications.

Future Improvements

During the development of the machine, some of the design choices were made based on monetary and time constraints. This resulted in a few less-than-optimal designs that could be modified in the future.

Carriage Stabilizers

The current design for the carriage stabilizers requires them to be hand tightened and does not provide the controller with state information. An option for the future would be to add an electric motor to activate them, or to replace them with some form of pneumatic clamps. In either case, both issues would be solved, and it would allow for more precise clamping.

Damping

The damping assembly could be integrated into the pneumatic system after performing additional analysis and/or the existing pneumatic activator that initiates the drop could be replaced. This could control the drop of the leaf such that it would stop before hitting the frame. This would reduce cost and machining hours and could improve the system performance, preventing the “kick back” of the leaf into the carriage after the drop.

Structure

The structure is currently the biggest source of machining hours, weight, raw material cost, and alignment issues within the assembly. New iterations would likely need to change the structure. Reexamining options like 8020 or equivalent materials could prove beneficial, although they would require full structural and impact analyses and involve a large redesign.

Electronics

The biggest future improvement for electronics would be to add a digital height readout. This was initially attempted with a linear encoder but the carriage movement during drops proved to be too violent for this type of sensor. A distance sensor could be mounted to the top of the frame and used to read the height from the carriage. This would allow for the code to fully control the height of the carriage and only allow height adjustments of 2”.

The electronics could be improved by using surface mount components to place the components directly onto the PCB, making it smaller, more robust to shock and vibrations, and improving performance.

Another important improvement would be to upgrade the wire connectors on the machine. The current connectors are JST connectors which, over time, have proved to not be as reliable as an industrial machine requires. They do not cause any safety or machine risks, but if one fails it will trigger an error and will need to be replaced.

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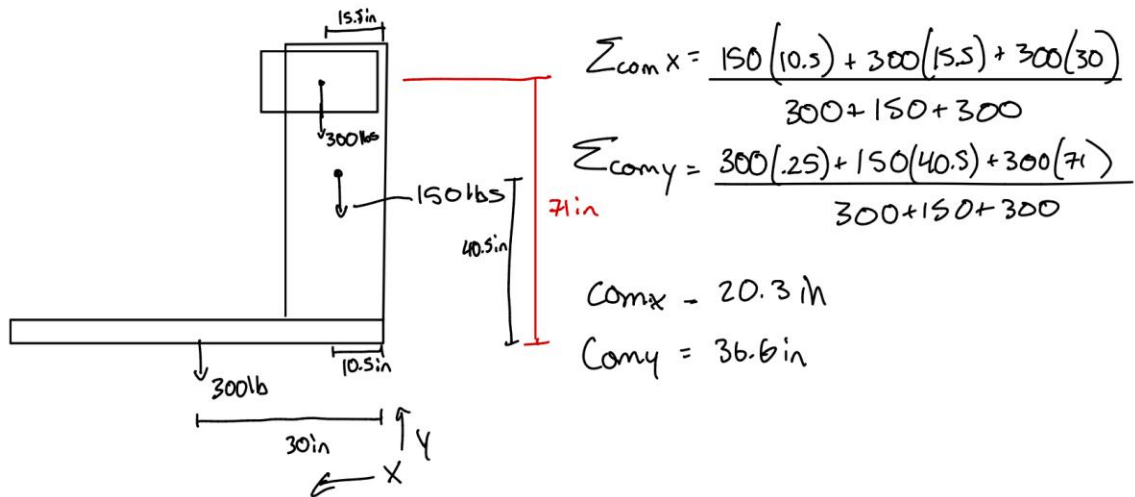
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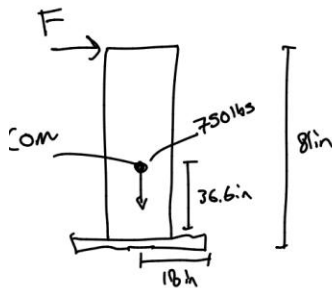
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Appendices

Appendix A – Center of Mass Calculations



Worst case



$$81 \text{ in } F = 750 \text{ lbs} \cdot 18 \text{ in}$$

Required $F = 166 \text{ lbs}$ laterally at highest point

Appendix B – Drop Leaf Geometry Calculations

